How much of a flood model's error can roughness explain? Adjoint-based measurement and calibration in E3SM's river dynamical core

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Abstract

We present what is, to our knowledge, the first *transient* discrete adjoint in a component of the Energy Exascale Earth System Model (E3SM): its river dynamical core, RDycore. Rather than rewrite the model in a differentiable-programming framework, we differentiated the production solver in place, keeping its discretization, data structures, and GPU backends. One gradient costs about one extra forward solve regardless of how many parameters are being calibrated, and every derivative is validated against finite differences of the full nonlinear solve.

We use the capability to ask a question calibration studies rarely ask: not how much error an optimizer removed, but how much of the model's error a parameter is *able* to remove. Tests with synthetic observations show that what a calibration can recover depends on where the observations are taken and what they measure, not on how many there are. Manning roughness is the parameter depth-averaged flood models usually calibrate, and for Hurricane Harvey on a 2.93M-cell Houston-area mesh, scored against 46 surveyed high-water marks, its ceiling can be measured before any optimizer runs: a single factor on the whole roughness field closes at most 0.08 m of a 0.72 m gap, and only at physically indefensible roughness values; fifteen land-cover classes calibrated separately close 0.107 m — still only 15% of the error. Sixteen forward runs (one per class, perturbed, plus a baseline) show why: the marks carry about *one* parameter's worth of independent information about roughness, so most of what a fifteen-class calibration reports comes from the prior, not from the data.

Pointing the same measurement at the model's initial state finds a larger effect: reducing the water stored in the domain at the start of the calibration window by 20% is worth 0.075 m, from a perturbation far easier to defend than the extreme roughness values needed to match it. Fitting an initial state of millions of unknowns to 46 observations would mean little, so we read the result as a diagnosis: most of the remaining error lives in the model's water balance, not in its friction.

A few forward runs are enough to establish this ceiling, and reporting it alongside the error an optimizer actually removed shows how much of a model's error was ever the parameter's to remove.

1 Introduction

Manning roughness is the dominant tunable parameter of depth-averaged flood models, and its calibration is traditionally manual or ensemble based. Adjoint methods deliver the gradient of an observation misfit with respect to *every* cell's roughness at the cost of about one extra forward solve, making distributed calibration tractable. The approach is well established in shallow-water modeling, from channel networks [1] and unstructured finite-volume *twin experiments* — calibrations against observations the model itself generated from a known roughness field, so that the recovered field can be scored against a truth, and unrelated to the "digital twin" of a physical asset [2] — to GPU-accelerated optimizers [3]. It is deployed in variational-assimilation platforms such as DassFlow [4, 5, 6] and the Thetis coastal model, which calibrated a spatially varying Manning field for the Baltic Sea with an automatically generated adjoint [7], following storm-surge sensitivity studies that addressed the overfitting inherent in distributed friction estimation [8]. The closest published work to ours is Pujol et al. [9], who infer distributed friction in a 2D urban flood model by variational assimilation of multi-source data *including high-water marks*, and use adjoint-derived global sensitivity measures to decide how to spatialize the friction field before assimilating — the same observable and the same design move that Sections 7.1 and 7.2 employ, at street scale and with a Tapenade-generated adjoint of a solver built for assimilation. What we add is neither the adjoint nor the observable but the setting and the question: a production Earth-system component differentiated in place, and a measurement of how much of the model's error the parameter can explain at all. The measurement is not specific to roughness, and Section 8 turns it on the model's initial state — where it finds more.

A recent generation of *differentiable* flood and hydrology codes obtains gradients by rewriting simplified solvers inside machine-learning frameworks: Inunda (PyTorch, local-inertial, calibrated on a Hurricane Harvey hindcast) [10], Hydrograd (Julia) [11], AegirJAX [12], CaMa-Flood-GPU [13], and differentiable routing [14]. These demonstrate demand for exactly this capability, but none is a production Earth-system component: they trade the validated numerics, unstructured meshes, and coupling infrastructure of codes like RDycore for differentiability.

This paper takes the complementary route: make the production code differentiable. That route has a long precedent outside E3SM — most prominently MITgcm, whose exact adjoint is generated by automatic differentiation of the production ocean model [15] and underpins two decades of ECCO state and parameter estimation [16]. Ours differs in how the derivative is obtained rather than in ambition: the Jacobian is hand-differentiated and assembled, which makes it available to the implicit solver as well as to the adjoint, and Section 5 argues that for a differentiable friction term the implicit solve is not optional. Inside E3SM the precedent is narrower: the land-ice component MALI [17] routinely inverts for basal friction with an adjoint obtained by automatic differentiation of its Albany velocity solver [18] — but that is the adjoint of a *steady* nonlinear solve, used for initialization; no E3SM component has, to our knowledge, a discrete adjoint of its transient time stepper. RDycore is a finite-volume shallow-water dynamical core (Roe fluxes, well balancing, wetting and drying) validated at up to 471M cells. We add the missing derivative infrastructure — an exact assembled Jacobian and TSAdjoint-based sensitivities [19] — entirely behind a configuration flag, with no impact on existing runs and no new dependencies.

Contributions: (i) an exact, FD-verified Jacobian of the SWE right-hand side, including Roe dissipation, entropy-fix branches, wet/dry branches, and reflecting/Dirichlet/critical-outflow boundary conditions; (ii) adjoint gradients dJ/du_0 and dJ/dn with respect to water-height observations at gauge locations at multiple observation times, FD-validated to 10^{-5} or better; (iii) TAO-based recovery of per-region, per-cell and land-cover-class roughness fields, with a quantified study of when the observing network makes them identifiable at all; (iv) implicit (backward Euler) stepping and its adjoint, enabled by the same Jacobian, with an empirical demonstration that explicit friction fails at operational time steps on a Hurricane Harvey configuration; and (v) a catalog of the non-differentiable structures encountered — useful, we believe, to anyone retrofitting adjoints onto an operational flood code.

2 Forward model

RDycore solves the 2D shallow water equations in conservative variables $u = (h, q_x, q_y)$, $\mathbf{q} = h\mathbf{v}$:

$$\partial_t h + \nabla\cdot\mathbf{q} = Q_r(x,t), \tag{1}$$

$$\partial_t \mathbf{q} + \nabla\cdot\left(\frac{\mathbf{q}\otimes\mathbf{q}}{h} + \frac{gh^2}{2}\,\mathbb{I}\right) = -\,gh\nabla z_b \;-\; g\,n^2 h^{-7/3}\,\mathbf{q}\,\|\mathbf{q}\|, \tag{2}$$

with cell-centered finite volumes on an unstructured mesh:

$$\frac{du_i}{dt} = -\frac{1}{|\Omega_i|}\sum_{e\in\partial\Omega_i} \mathcal{F}\big(u_i, u_j, \mathbf{n}_e\big)\,L_e + S(u_i; n_i, t) \equiv f_i(u, n), \tag{3}$$

where $\mathcal{F}$ is the Roe flux with a critical-flow (entropy) fix, primitives are reconstructed with a regularized division and a dry cutoff $h_{\min}$, and S contains bed slope, Manning drag, and external forcing.

The Roe flux's dissipation is built from the moduli $|\lambda_p|$ of its characteristic wave speeds, so the closed-form expression it evaluates switches wherever a wave speed changes sign. Switches like this — *branches*, in the sense that $|x|$ has a positive and a negative branch — recur throughout the discretization, and they belong to the model rather than to the derivative: a nonlinear solver must accommodate them whether or not anyone differentiates the code, which is the argument of Section 5, and Section 5.1 catalogs the ones this work encountered. Away from a switching surface the map is smooth and equals its active branch; the derivative machinery of Sections 3.1–3.2 is *branch-faithful*, differentiating the expression the condition actually selected to give the correct one-sided derivative.

A differentiable right-hand side. A prerequisite that deserves emphasis: RDycore's existing friction treatments (a semi-implicit splitting and the implicit scheme of Xia and Liang) embed the time step — and, in the semi-implicit case, the flux divergence — *inside* the RHS callback. The resulting map is consistent only with forward Euler and admits no well-defined $\partial f/\partial u$. We therefore added a plain explicit source treatment (`source.method: explicit`), used on every adjoint path; Section 5 shows how the resulting stiffness is then handled properly, by the implicit integrator rather than inside the explicit right-hand side f.

3 Differentiating the production solver

Two design decisions shape everything downstream. The derivative of the right-hand side is obtained by hand — forward-mode differentiation of the flux computation itself, branch by branch

— and it is *assembled* into a sparse matrix rather than applied matrix-free. Assembly is what lets one artifact serve three consumers: the implicit solver preconditions with it (Section 5), `TSAdjoint` transposes it, and the calibration couples it to the parameter Jacobian. This section builds that Jacobian (Section 3.1) and the adjoint sensitivities on top of it (Section 3.2); every derivative is gated by finite differences of the full nonlinear solve.

3.1 Exact Jacobian

The RHS Jacobian is block sparse with 3×3 blocks on the cell-adjacency graph: each interior edge (i, j) contributes $\mp(L_e/|\Omega_{i,j}|)\,\partial\mathcal{F}/\partial u_{L,R}$ to the two cell rows, local sources contribute diagonal blocks, and boundary edges contribute diagonal blocks through the ghost-state map of the boundary condition.

Flux blocks by forward-mode differentiation. Rather than deriving the dissipation-matrix derivatives symbolically, we differentiate the flux *computation*: a hand-written forward-mode (tangent) version of the Roe flux routine propagates directionals through every intermediate — Roe averages, eigenvectors, eigenvalues including both branches of the critical-flow fix, wave strengths, and physical fluxes — taking the code's own branch at every $|\cdot|$ and switch (a consistent one-sided derivative). Seeding unit directions through the primitive-reconstruction chain rule yields the two conservative-space blocks in six tangent evaluations per edge. The approach mirrors the flux code line by line, which makes it reviewable and, in our experience, correct on first assembly against the verification harness below.

Boundary blocks. For a boundary edge, $F(u_{\text{in}}, u_g(u_{\text{in}}))$ contributes $(\partial_L F + \partial_R F\,\partial u_g/\partial u_{\text{in}})$. The reflecting map is a linear mirror; Dirichlet ghosts are constant; critical outflow has a closed-form ghost map $h_g = (q^2/g)^{1/3}$ with the code's inflow branch producing an identically zero flux.

Parameter Jacobian. Manning n appears only in the drag term, so $\partial S_{\mathbf{q}}/\partial n = -2gnh^{-7/3}\mathbf{q}\|\mathbf{q}\|$: two nonzeros per cell, supplied to `TSSetRHSJacobianP`.

Verification. Three layers, all against central finite differences, with the pass/fail tolerances fixed in advance of the runs (Table 1): an edge-level harness that differences the flux routine itself on sampled states (with a Richardson self-check of the FD order); full-matrix comparisons of the assembled Jacobian against a matrix-coloring FD Jacobian of the actual RHS on three state/boundary combinations, boundary rows included; and end-to-end gradient checks through the adjoint (next section). A finite-difference coloring Jacobian (`numerics.jacobian: fd`) remains available permanently as the verification baseline.

3.2 Adjoint sensitivities

For gauge observations y_k at times t_k with observation operator H (one row per gauge, selecting water height) and error variance σ^2,

$$J(n) = \tfrac{1}{2\sigma^2} \sum_k \|Hu(t_k) - y_k\|^2. \tag{4}$$

`TSAdjoint` [19] differentiates the discrete time stepper, so gradients are exact to roundoff for the computed trajectory — the property that makes the FD gates in Table 1 meaningful. Multi-time observations are handled by a segmented backward sweep: the adjoint is advanced between

Check	Configuration	Result	Gate
Edge flux blocks	wet jump 10 m/5 m	7.6×10^{-9}	10^{-6}
Edge flux blocks	transcritical (entropy-fix branch)	2.4×10^{-10}	10^{-5}
Source block	shallow flowing state	$< 10^{-6}$	10^{-6}
Assembled matrix	uniform flow, reflecting BCs	1.6×10^{-8}	10^{-6}
Assembled matrix	dam break, reflecting BCs	1.5×10^{-8}	10^{-6}
Assembled matrix	dam break, Dirichlet + critical outflow	1.9×10^{-8}	10^{-6}
dJ/du_0 (explicit RK)	8 sampled components	1.0×10^{-8}	10^{-5}
dJ/dn (explicit RK)	domain aggregate	2.9×10^{-6}	10^{-5}
dJ/du_0 (implicit BE)	tight inner tolerances	1.4×10^{-8}	10^{-5}
dJ/dn (implicit BE)	tight inner tolerances	5.3×10^{-6}	10^{-5}

Table 1: Relative errors of analytic derivatives against central finite differences of the full nonlinear computation. Every check in this table is an automated test in RDycore's `ctest` suite, run on every change to the repository.

observation times with `TSAdjointSetSteps` and receives a jump $H^T\sigma^{-2}r_k$ at each t_k (Algorithm 1 in Section 4). One sweep yields both dJ/du_0 (state sensitivity) and dJ/dn (per-cell parameter gradient, via the two-nonzero parameter Jacobian).

Two integrator facts constrain the design. PETSc's plain forward-Euler integrator implements no adjoint, so adjoint-path configurations use the Runge–Kutta family. And for implicit integrators the discrete-adjoint gradient is only as exact as the inner solves: the Manning gradient fails its gate at default inner tolerances and passes once they are tightened (Table 1, implicit rows). Loose inner tolerances also turned out to be a false economy on this problem: a tighter Krylov tolerance makes the implicit forward *faster*, because better linear solves let Newton converge in fewer iterations, and the production setting (Krylov 10^{-4}, nonlinear 10^{-5}) reproduces the fully converged answer to every printed digit for about a quarter more wall time than the loosest usable setting. Campaigns therefore run at converged inner tolerances, and the verified gradients *are* the production gradients. What the FD gates cannot certify at production scale is a different matter entirely — the objective's own nonsmoothness over long windows — quantified in Section 7.1.

4 The calibration algorithm: strong-constraint 4D-Var

The complete calibration loop is summarized in Algorithms 1–3. It is a strong-constraint 4D-Var iteration: the model is enforced exactly (the only controls are the Manning field and, optionally, the initial state). Each objective evaluation performs one checkpointed forward sweep that records the gauge residuals r_k at the observation times, followed by one backward sweep of the discrete adjoint (Algorithm 2). The per-iteration cost is therefore one forward plus one adjoint solve, independent of the number of parameters. Throughout, the objective splits as $J = J_{\mathrm{mis}} + J_{\mathrm{prior}}$, where the *misfit* $J_{\mathrm{mis}} = (2\sigma^2)^{-1}\sum_k \|r_k\|^2$ measures disagreement with the observations and J_{prior} is the Tikhonov term of Eq. (5). Only J_{mis} needs the adjoint; the prior term's gradient is closed form.

The backward sweep carries two vectors with distinct roles. The *adjoint state* λ holds $\partial J_{\mathrm{mis}}/\partial u(t)$ at the current point of the backward traversal. It is seeded with the terminal observation sensitivity $\sigma^{-2}H^T r_K$, propagated one step at a time by the transposed Jacobian of the forward step map (the backward "model step"), and incremented by a jump $\sigma^{-2}H^T r_k$ each time the sweep crosses an observation time, because the state at that time enters the misfit directly. The *parameter sensitivity* μ starts at zero and accumulates, at every step, the chain-rule contribution $(\partial\Phi_s/\partial n)^T\lambda$ of that step's Manning dependence. When the sweep reaches $t = 0$, μ is the complete misfit gradient dJ_{mis}/dn

and λ is dJ_{mis}/du_0. Both transposed Jacobians are the analytic operators of Sections 3.1–3.2 — the assembled RHS Jacobian for the state propagation, the two-nonzero parameter Jacobian for the μ accumulation — applied within `TSAdjoint`'s discrete adjoint of the chosen integrator.

The outer update is TAO's bound-constrained limited-memory quasi-Newton method BLMVM [20] (Algorithm 3). The loop terminates on the standard bound-constrained test: the projected gradient g^P — the gradient with components pointing out of the feasible box zeroed — drops below an absolute tolerance, or an iteration budget is reached. The same loop serves every parameterization in this paper: per-region, per-cell, and land-cover-class fields differ only in the expansion of the parameter vector onto cells (line 3 of Algorithm 1) and the corresponding summation of μ back onto parameters.

Algorithm 1 Strong-constraint 4D-Var calibration of the Manning field, as implemented with TAO/BLMVM and `TSAdjoint`. The window map $M_{t_{k-1}\to t_k}$ is the composition of m steps of the discrete forward stepper (explicit RK, backward Euler, or ARK-IMEX).

Require: prior/initial guess n_0, bounds $[n_{\min}, n_{\max}]$, observations y_k at times $t_k = k\,m\,\Delta t$ ($k = 1, \dots, K$), gauge operator H, error σ, Tikhonov weight β, gradient tolerance ε_g, iteration budget $I_{\max}$

1: $n \leftarrow n_0$
2: **repeat**
3: set the model's cellwise drag coefficient from n ▷ identity for per-cell n; constant over each region or land-cover class for the reduced parameterizations
4: $u \leftarrow u_0$ ▷ fixed initial state (strong constraint)
5: **for** $k = 1, \dots, K$ **do** ▷ forward sweep; trajectory checkpointed — every step, or a revolve schedule for long windows (Section 5)
6: $u \leftarrow M_{t_{k-1}\to t_k}(u;\, n)$
7: $r_k \leftarrow Hu - y_k$
8: **end for**
9: $J \leftarrow \dfrac{1}{2\sigma^2} \sum_{k=1}^{K} \|r_k\|^2 + \dfrac{\beta}{2}\|n - n_0\|^2$
10: $(\lambda, \mu) \leftarrow \textsc{AdjointSweep}(\{r_k\})$ ▷ Algorithm 2 differentiates the stepper above, so $\mu = dJ_{\text{mis}}/dn$ is exact to roundoff and inner-solver tolerance (Section 3.2)
11: $g \leftarrow \mu + \beta\,(n - n_0)$
12: $n \leftarrow \textsc{BlmvmStep}(n, J, g)$ ▷ Algorithm 3
13: **until** $\|g^P\| \le \varepsilon_g$ or $I_{\max}$ iterations
14: **return** n

Practical execution. One objective evaluation costs one checkpointed forward sweep plus one adjoint sweep, independent of the number of parameters; each trial step of the BLMVM line search costs one such evaluation. A basin-scale window needs more iterations than a single batch allocation provides, so a run writes its parameter vector on exit and the next reads it back as its starting iterate, matched by land-cover class rather than by position. The resume is exact in the parameters — the continuing run's first objective reproduces the previous run's last to six digits — but BLMVM's stored curvature pairs do not survive it, so a chain of jobs is not step-for-step identical to one long run. On a twin at the Houston mesh the restart was not a penalty: a chained $2+2$ iteration run reached a lower objective than a single cold four-iteration run, the restart acting as a rescaling of the quasi-Newton model. That is one twin, not a rule; the point for planning is that chaining did not cost convergence there.

Algorithm 2 AdjointSweep: segmented discrete-adjoint backward sweep (`TSAdjointSolve` with `TSAdjointSetSteps`). Φ_s denotes the forward map of step s, $u_s = \Phi_s(u_{s-1}; n)$; its transposed Jacobians are built from the assembled RHS Jacobian $\partial f/\partial u$ (Section 3.1) and the parameter Jacobian $\partial f/\partial n$ according to the integrator's discrete-adjoint formula.

Require: checkpointed trajectory $\{u_s\}_{s=0}^{Km}$, residuals $\{r_k\}_{k=1}^{K}$ at steps $s = km$, gauge operator H, error σ

Ensure: $\lambda = dJ_{\text{mis}}/du_0,\ \ \mu = dJ_{\text{mis}}/dn$

$\lambda \leftarrow \sigma^{-2} H^T r_K$ ▷ λ holds $\partial J_{\text{mis}}/\partial u(t_s)$ throughout

$\mu \leftarrow 0$

for $s = Km, \dots, 1$ **do**

  restore u_{s-1} (and stage values): read the checkpoint, or recompute forward from the nearest one when the checkpoint schedule demands

  $\mu \leftarrow \mu + \big(\partial\Phi_s/\partial n\big)^T \lambda$ ▷ accumulate this step's Manning dependence

  $\lambda \leftarrow \big(\partial\Phi_s/\partial u_{s-1}\big)^T \lambda$ ▷ backward model step (transposed step Jacobian)

  if $s - 1 = km$ for some $1 \le k < K$ **then**

    $\lambda \leftarrow \lambda + \sigma^{-2} H^T r_k$ ▷ observation jump: u_{km} enters J_{mis} directly

  end if

end for

return (λ, μ)

Algorithm 3 BlmvmStep: one iteration of TAO's bound-constrained limited-memory variable-metric method [20]. $P_{[n_{\min}, n_{\max}]}$ is componentwise clipping to the bound box; the curvature pairs $\{(s_i, y_i)\}$ persist across calls. The projected gradient used in the convergence test of Algorithm 1 is $g_i^P = g_i$ for n_i strictly inside the box, $\min(g_i, 0)$ at the lower bound, and $\max(g_i, 0)$ at the upper bound.

Require: iterate n, objective J, gradient g, stored pairs $\{(s_i, y_i)\}$, bounds $[n_{\min}, n_{\max}]$

$d \leftarrow -B\, g^P$ ▷ $B \approx (\nabla^2 J)^{-1}$ by the L-BFGS two-loop recursion over the stored pairs

find a step length $\tau > 0$ with a projected line search on $n(\tau) = P_{[n_{\min}, n_{\max}]}(n + \tau\, d)$ satisfying sufficient decrease of J ▷ each trial τ costs one forward-plus-adjoint evaluation (lines 3–11 of Algorithm 1)

$n^+ \leftarrow n(\tau)$; evaluate g^+ at n^+

$s \leftarrow n^+ - n$; $y \leftarrow g^+ - g$; store (s, y) if $s^T y > 0$, discarding the oldest pair beyond the memory limit

return n^+

Scaling the optimization problem. A quasi-Newton method needs its variables and its objective to be $O(1)$, and in physical Manning units this problem is not: on the production window $\|g\| \approx 1.5 \times 10^3$ while $n \approx 0.05$. Starting from $B = I$, the first trial point $n - g$ therefore projects every component onto the lower bound — exactly where the model cannot solve, since a uniform $\alpha = 0.2$ diverges (Table 7) — so every line-search trial fails and the optimizer returns having taken no step at all. The failure is worth recording because of how it presents: nothing is wrong with the gradient, and the finite-difference gates pass throughout.

Two changes make the problem well scaled, and both have physical readings rather than being tuning constants. First, optimize the *relative* field $\alpha_k = n_k/n_{\text{prior},k}$ and the relative objective $J/J(n_{\text{prior}})$. Here and throughout, α is a multiplier on roughness, never a step length: a class at $\alpha = 1.5$ carries 1.5 times its lookup value, and the uniform scans of Section 7.2 set every class to the same α. Both are then $O(1)$ and the unit quasi-Newton step becomes a roughness change of order 10%: at the NLCD start point of the production configuration, over a shortened window that reproduces the zero-step failure exactly, $\|g\|/J_0 = 0.10$, and the same start point that took no step in physical units takes one that lowers the objective and the peak-WSE MAE together. The bounds become statements about relative roughness, $\alpha \in [0.3, 3]$, which additionally excludes the region where the implicit solve fails. Second, set the Tikhonov weight from a prior standard deviation on Manning n rather than by hand,

$$J = \underbrace{\frac{1}{2\sigma^2}\sum_k \|r_k\|^2}_{J_{\text{mis}}} + \underbrace{\frac{\beta}{2}\|n - n_{\text{prior}}\|^2}_{J_{\text{prior}}}, \qquad \beta = \sigma_n^{-2}, \tag{5}$$

and report σ_n. This matters because β is dimensional: a bare value is uninterpretable across problems, and ours had been chosen independently in every experiment, spanning six orders of magnitude. Against the measured $J(\alpha)$ curve, $\sigma_n = 0.015$ places the minimum of the uniform-scaling direction at $\alpha \approx 0.70$, while $\sigma_n = 0.020$ leaves no interior minimum there at all — the same pathology the hand-chosen weight had. The prediction is measured: the scan point at $\alpha = 0.70$ was run afterwards and returned MAE 0.6894 against the 0.6895 the balance implied (Table 7). Stating the prior turns the choice into an arithmetic consequence of a quantity the domain team can argue about.

Absolute or fractional prior — not a cosmetic choice. Equation (5) states the prior as an absolute departure in n, and for a land-cover table that is the wrong statement. A single $\sigma_n = 0.015$ over an NLCD lookup spanning $n = 0.027$ (barren) to $n = 0.16$ (developed high) asserts a $\pm 56\%$ prior on one class and $\pm 9\%$ on another; and because the penalty scales as n_{prior}^2, a small-n class buys a large *fractional* excursion cheaply. The iterates show exactly that signature: at the production configuration's first quasi-Newton step, every class driven to the upper bound was small-n (pasture 0.038, developed-open 0.040, developed-low 0.090) and every class driven to the lower bound was large-n (developed-high 0.160, developed-medium 0.120, shrub 0.115, woody wetland 0.098) — an ordering by prior value, not by hydrology. Penalizing the fractional departure $n_k/n_{\text{prior},k} - 1$ with a width σ_α instead treats the classes alike, and asks the domain expert a question they can answer: how far can the lookup be wrong, in percent?

The tension survives the reformulation, and that is the substantive result rather than an implementation detail. Holding the first step's excursion requires $\sigma_\alpha \approx 0.15$: one must believe the NLCD values are good to $\pm 15\%$ for the calibration to stay physical, while at $\pm 30\%$ the data drives it somewhere no hydrologist would defend. *Within a prior wide enough to be honest, this observable cannot deliver a defensible distributed roughness field.* We report the prior width used with every number and regard its value as an open question for the domain team — it heads the list of questions at the end of this draft — not as a tuning parameter.

The diagnosis is worth separating from the fix, because the failure mode is not specific to this model: an optimizer that takes zero steps on a problem whose gradient passes every verification gate is a scaling defect, and the cheapest test is to compare $\|g\|$ against the scale of the variable before suspecting the derivative.

5 Implicit stepping, enabled and required

The explicit drag term is stiff in thin water: linearizing gives a stability limit $\Delta t \lesssim 2h^{4/3}/(gn^2|v|)$, which for sheet flow at centimeter depths is seconds — far below the gravity-wave CFL limit of kilometer-scale meshes. Our Hurricane Harvey test configuration (Houston at 1 km, $\Delta t = 30$ s) confirms this empirically: the explicit source treatment produces NaN within the first objective evaluation, in quantitative agreement with the analysis. The same holds at basin scale, where the margin is far wider: on the spun-up 30 m, 2.93M-cell Harvey mesh (99.9% wet, median depth 6 mm) the friction-rate limit is $\Delta t \lesssim 2.4$ ms, and forward Euler of the differentiable right-hand side returns a non-finite state at every tested step ($\Delta t = 0.25$, 0.125, 0.05 s) while backward Euler at $\Delta t = 1$ s completes a 72-hour window without a single Newton failure. There is no viable explicit path for the differentiable right-hand side at this resolution.

The assembled exact Jacobian makes the remedy nearly free: backward Euler via PETSc's theta method, with SNES driven by the same Jacobian, carries the Harvey window at $\Delta t = 30$ s, and the theta-method adjoint passes the same FD gates (Table 1).

We also implemented the sharper tool: an ARK-IMEX splitting with the Manning drag as the stiff implicit part (`temporal: arkimex`). The implicit stage system is block-diagonal — an independent 3×3 Newton per cell, no global solve — while fluxes and bed slope remain explicit under the gravity-wave CFL limit. Its adjoint passes the same gates (2.0×10^{-8} for dJ/du_0, 2.5×10^{-6} for dJ/dn) and it carries the Harvey window at $\Delta t = 30$ s at 6.7 s per forward-plus-adjoint sweep, with gradients matching backward Euler to three digits. The splitting also runs under RDycore's libCEED operator backend, whose explicit side uses friction-free source Q-functions; the two backends agree to machine precision (5×10^{-16} relative) after 100 ARK-IMEX steps. A device-resident implicit part is the natural follow-up.

Three implementation facts others will hit:

- PETSc's ARKIMEX adjoint dereferences *both* the implicit- and explicit-part parameter Jacobians, so a zero explicit-part `TSSetRHSJacobianP` must be registered even when the parameter appears only implicitly.
- The embedded step-size controller must be disabled for finite-difference gradient validation: perturbed solves otherwise take different step sequences, which shows up as an apparent $\sim 10^{-4}$ gradient error that vanishes with fixed steps.
- PETSc's in-memory checkpointing trajectory cannot tolerate a *segmented* forward solve (one `TSSolve` per observation window). `TSSolve` sets its converged reason before the final trajectory write of every call, which the memory trajectory interprets as the end of the forward run, corrupting the checkpoint schedule at every interior observation time. The forward sweep must be a single `TSSolve` with observations recorded by a monitor; the segmented backward sweep is unaffected.

Rain-forced configuration (land-cover prior)	Outlet	Outcome
600-step control	critical outflow	Newton pinned from step 498
600-step control	free outflow	zero failures; 598/600 solves in two iterations
1-h calibration window (forward + adjoint + line search)	free outflow	clean end to end
72-h forward through the Harvey crest (2.93M cells)	free outflow	259,200 solves, zero failures

Table 2: The outlet replacement is a cure, not an improvement. Under critical outflow, tolerance relief bought steps only linearly; the transmissive outlet completes the identical window and every longer one attempted.

Newton robustness is set by residual discontinuities, not by conditioning. Where implicit stepping fails on this problem, it fails for a reason that is invisible in the smooth twin experiments: the residual is discontinuous, so Newton has nothing to converge *to*. At the land-cover roughness prior, rather than the twins' smooth low-n fields, backward Euler descends cleanly for a dozen iterations and then freezes — a vanishingly small line-search step changes the residual norm severalfold, the signature of a jump, not of ill-conditioning. Two mechanisms are responsible, both catalogued in Section 5.1 and both invisible until the parameter field is realistic: the drag's wet/dry cutoff, whose amplitude scales as n^2, and the critical-outflow boundary's stagnation switch.

Both fixes are implemented, gated, and opt-in (open issues 4–5). Regularizing the drag velocity removes the first and restores honest convergence over the early transient. The second we removed by replacing the outlet condition itself: a transmissive (zero-gradient) free outflow — the ghost state copies the interior state, the convention of most finite-volume shallow-water codes — is continuous in the interior state by construction, and its ghost map is the identity, so the analytic Jacobian and adjoint carry it exactly, FD-gated like every other block. The effect is not an improvement but a cure (Table 2): the identical failing control completes with zero nonlinear failures, and the fix holds through the longest windows we run. Before the replacement we absorbed the pin by loosening the nonlinear tolerance; a class-twin calibration under that crutch still recovered all fifteen class values to machine precision, so the gradient path was unimpaired — tolerable, but a crutch. The general lesson for adjoint retrofits is that nonsmoothness in an operational flood code is not only a differentiability question — it is a nonlinear-solver question.

Checkpointed adjoints for long windows. A naive stored trajectory for a full 36-hour Hurricane Harvey window at the 30 m mesh's CFL-limited $\Delta t = 0.25$ s (518,400 steps, 2.93M cells) would occupy tens of petabytes. PETSc's checkpointing trajectory (`-ts_trajectory_type memory` with the revolve optimal-schedule library) replaces storage with recomputation. We validated it across the full hierarchy. On the dam-break twin it reproduces the disk-trajectory calibration and gradient gates bit-for-bit for explicit RK, backward Euler, and ARK-IMEX (the implicit integrators need the same tight inner tolerances as in Section 3.2, since checkpoint recomputation re-solves the stage systems). On the 30 m mesh, a forward-plus-adjoint sweep with only *three* states checkpointed in memory matches the disk-trajectory gradient to all printed digits — and runs slightly faster, because recomputation is cheaper than trajectory I/O at this scale. For the full window, a few hundred checkpoints ($\sim$14 GB aggregate) bound the recompute overhead by a factor of about four, making the long-window adjoint memory-feasible; the remaining cost is the forward compute itself, addressed next.

5.1 Nondifferentiability catalog

[Reviewers: do we want this subsection at all? It is the least conventional part of the paper. Five of its six entries are now resolved with gated fixes, which is an argument for keeping it as the record of what retrofitting an adjoint onto an operational flood code actually costs — and the counter-argument is that a resolved catalog is engineering detail that belongs in an issue tracker. Cut, keep here, or move to an appendix.]

Retrofitting an adjoint onto an operational flood code surfaces nonsmooth structure worth documenting. Every entry below is annotated with its status: five are resolved and gated in the test suite, and one remains open.

- **Time step inside the RHS.** *(Resolved.)* Operator-split friction treatments (here: semi-implicit and XQ2018) make f depend on Δt and on intermediate flux divergences; no consistent $\partial f/\partial u$ exists. The remedy is an explicit f plus a genuinely implicit integrator, not a smarter derivative — which is the architectural choice this section exists to justify, and the reason the implicit-versus-explicit cost comparison above is not the one a forward-only code would make.

- **Entropy-fix branches.** *(Resolved.)* The critical-flow fix is piecewise smooth; branch-faithful one-sided derivatives pass FD checks even on transcritical states (Table 1).

- **Critical-outflow stagnation.** *(Resolved by changing the boundary condition.)* The outflow ghost map $h_g = (q^2/g)^{1/3}$ has an unbounded one-sided derivative at $q = 0$, and the inflow branch switches there: a genuine subgradient point. Unlike the others in this list it is not merely a derivative subtlety — the branch switch is a discontinuity of the *residual* itself, since the $u_\perp < 0$ branch zeroes both states (a wall) while the $u_\perp \geq 0$ branch imposes the critical ghost, so the edge flux jumps by the full wet-onto-dry Roe flux, $O(gh^2/2)$, as a boundary cell crosses stagnation. With an implicit integrator this stalls Newton outright: the line search collapses to $\lambda \sim 10^{-6}$ with the residual pinned at a floor of 10^{-4}–10^{-5} relative. Explicit stepping never sees it. The cure is to change the boundary condition, not the solver: a transmissive (zero-gradient) free outflow — ghost state equal to the interior state, the standard free outflow of finite-volume shallow-water codes — is continuous by construction, has an identity ghost map in the Jacobian, and removes the stall entirely. Provided as an opt-in condition alongside the original, and gated: a 72-hour rain-forced basin forward completed 259,200 implicit solves with zero nonlinear failures.

- **Drag at the wet/dry cutoff.** *(Resolved by regularizing the drag velocity.)* The drag $gn^2h^{-7/3}\mathbf{q}\|\mathbf{q}\|$ gated at $h_{\min}$ with a plain $\mathbf{q}/h$ velocity is likewise residual-discontinuous: a cell crossing the cutoff with retained momentum switches an $O(h^{-7/3})$ term on and off. The amplitude scales with n^2, so it is invisible in smooth-n twin experiments and disabling at land-cover roughness. Evaluating the drag with the ANUGA-regularized velocity $\mathbf{u} = \mathbf{q}h/(h^2 + h_{\text{anuga}}^2)$ removes it (the drag then vanishes as $h^{5/3}$), and reduces identically to the plain form at $h_{\text{anuga}} = 0$. The value of h_{anuga} itself is an empirical choice rather than a physical one (open issue 5).

- **Wet/dry branches.** *(Resolved in the derivative; measured as objective nonsmoothness at basin scale.)* That a moving wet/dry front is the awkward part of a shallow-water adjoint is well known: Funke et al. built an adjoint wetting and drying scheme precisely to make derivative-based optimization usable for inundation problems [21]. In our setting the RHS's dry cutoffs are honored branch-faithfully in every block; fully wet validation cases keep FD comparisons clean, and dry cells are correctly insensitive to their roughness. At basin scale the branches also make the *objective* nonsmooth: over long rain-forced windows the trajectory crosses so many per-cell depth-cutoff surfaces that a finite-step central difference measures a smoothed slope rather than the one-branch derivative the adjoint returns — quantified, with the verification policy it implies, in Section 7.1.

- **The argmax in the peak-WSE observable.** *(Open — the one entry we have not closed.)* The high-water-mark misfit (Section 7.1) depends on the state through $\max_k Hu(t_k)$: piecewise smooth, with kinks where two sampled times tie for a mark's peak. The adjoint injects each mark's residual at its recorded argmax time, which is the exact gradient away from ties (a measure-zero set); but the peak time *moves* across optimizer iterates, so the objective the quasi-Newton method sees is only piecewise smooth and its line search can straddle kinks. Practice is untroubled — the FD gates pass away from ties and the twin calibrations descend monotonically — and the instrumented basin-scale gate adds a measurement: across every FD probe on the real 108-mark configuration, *no* mark's argmax moved and none flipped wet/dry, so at this geometry the argmax kinks are far sparser than the dynamics' wet/dry kinks (previous entry). A second, independent measurement points the same way: evaluating the production window on 4, 8 and 16 MPI ranks — three different domain decompositions, hence three different reduction orders — reproduces the peak-WSE objective and MAE to every printed digit ($J = 8.082566 \times 10^2$, MAE 0.7188 m), so no mark's recorded peak time is sensitive to partitioning at the level roundoff perturbs it. This remains the standard footing of max-type observables in calibration rather than a theorem. *Proposed remedy, not yet implemented:* freeze the argmax assignment across an outer iteration, so each inner subproblem is genuinely smooth in the parameters, and re-evaluate the peak times between outer iterations — a fixed point on peak times, converged when the assignment stops changing. This costs one extra forward per outer iteration and makes the smoothness assumption the line search relies on true by construction rather than true in practice. A smooth surrogate (softmax over the sampled times, tempered toward the true maximum) is the alternative, at the cost of biasing the peak it calibrates against.

6 Twin experiments: what the observing network can recover

In a twin experiment the observations are model output generated from a roughness field we chose, so a calibration can be scored against a known truth rather than against a real survey whose own errors are unknown. The per-cell twin runs on the Houston 1 km Hurricane Harvey mesh (2,746 cells, 4200 s window at $\Delta t = 30$ s). Truth is a two-zone field split at the domain midline, observed at 343 synthetic gauges over 14 observation times, calibrated with implicit stepping and Tikhonov regularization ($\beta = 10^{-5}$). Every cell is wet and moving in this configuration, so all 2,746 parameters are observable. The configuration is deliberately gauge-sparse — 343 gauges for 2,746 parameters, an 8:1 ratio — and it exhibits the classic *semiconvergence* of underdetermined inverse problems (Figure 1). At 300 BLMVM iterations ($\sim$22 minutes on one laptop core, $\sim$4.4 s per forward-plus-adjoint) the recovered map shows the two-zone truth structure cleanly at 26.3% relative L^2 error. Continuing the same weakly regularized optimization to 2000 iterations *degrades* recovery to 38.0%: the misfit keeps decreasing while the parameters drift along observation-null directions, ultimately pinning cells against both bound constraints (Figure 1, center left). Early stopping was acting as the effective regularizer. The behavior disappears when the problem is made well posed: a properly regularized Houston run ($\beta = 10^{-4}$, 686 gauges) reaches 19.4% at a 1000-iteration budget with no drift (Figure 1, center right) — the sharpest recovery of the three, with

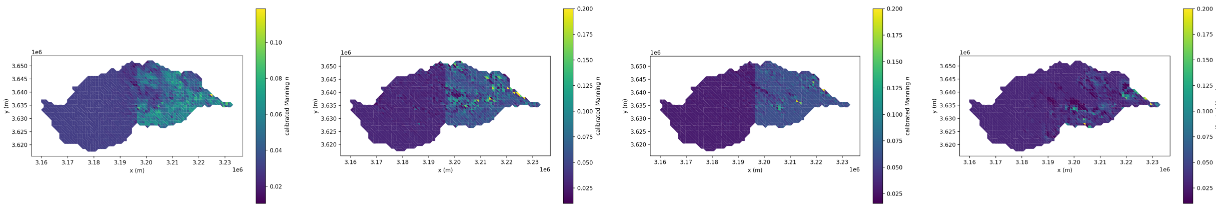

Figure 1: Regularization, semiconvergence, and network density on the Houston twin (2,746 parameters; truth: $n = 0.03$ west, $n = 0.06$ east of the midline; implicit stepping, 14 observation times). Far left: weak regularization ($\beta = 10^{-5}$, 343 gauges) stopped early at 300 iterations — 26.3% error. Center left: the same setup run to 2000 iterations — the misfit is lower but the field drifts along unobserved directions to 38.0% error and saturates the bounds $[0.01, 0.2]$. Center right: proper Tikhonov weight ($\beta = 10^{-4}$, 686 gauges), 1000 iterations — 19.4% error, no drift, sharp zone boundary. Far right: the *real* network — the 17 in-mesh USGS gauges with Harvey stage records, same $\beta = 10^{-4}$, 1000 iterations — the misfit falls 2×10^7-fold while the field recovers to only 44.9%: no trace of the two-zone truth, near-prior almost everywhere with null-space structure pinned against both bounds.

the zone boundary resolved at the truth midline. This is precisely the overfitting risk documented for distributed friction estimation [8], here reproduced in a controlled twin setting. It motivates the regularization studies (total variation, land-cover priors, discrepancy-principle stopping) planned as follow-on work.

The real gauge network. The synthetic gauge sets above are a controlled ramp; the network that matters is the real one. Twenty USGS gauges fall inside the Houston mesh, seventeen with Hurricane Harvey stage records.[1] Re-running the same twin with the observation operator built at those seventeen gauge cells — 238 observations (17 gauges × 14 times) against 2,746 parameters — is a sobering experiment. BLMVM drives the misfit down by a factor of 2×10^7, an essentially perfect fit of the observations, while the recovered field is 44.9% from the truth and saturates both bounds (Figure 1, far right). This is semiconvergence in its purest form: the real network's null space is so large that data fit says almost nothing about parameter recovery, and no Tikhonov weight or stopping rule rescues a per-cell field from 238 observations. The practical conclusion for real-data calibration is to shrink the parameter space to match the network — but, as the next subsection measures, shrinking the parameter space is necessary and not sufficient: whether a reduced parameterization is identifiable depends on the observation geometry and window, not on the ratio of observation to parameter counts.

6.1 Identifiability is a property of the observable: gauges versus high-water marks at 30 m

The natural reduced parameterization for this domain is one Manning value per NLCD land-cover class (15 classes at the 30 m Turning mesh, 2.93M cells), with the published NLCD lookup as prior and truth for twins. Counting is deceptively favourable: even the sparse real network supplies an order of magnitude more observations than parameters. Measurement says otherwise. In a class twin on the 30 m mesh at the real 13-gauge geometry (260 observations, a short early-transient

[1] HydroShare DOI `10.4211/hs.c037167e497546a1bc1508dfb32a9cff`.

window), the objective falls 128× while the recovered class values remain 66% wrong in relative L^2, worst class 81% — and the failure is structural: classes whose cells are dynamically distant from every gauge (the three forest classes) never move from the starting value, because within the window nothing their roughness does can reach a gauge. The identical configuration observed at 418,076 strided cells recovers all fifteen classes to machine precision, so the failure is the network, not the method. Fifteen parameters, thirteen gauges, and an "overdetermined" observation count: still unidentifiable.

The same question re-asked with the peak observable comes out differently. Observing the same NLCD-truth twin at the 108 QC-passed high-water-mark cells (peak WSE sampled over a one-hour rain-forced window, implicit stepping, free-outflow outlet; Section 7.1) and calibrating fifteen classes from a uniform start: the objective falls 81×, the peak-WSE error at the marks drops from 0.103 m to 0.017 m MAE, and the area-dominant classes are identified — the three developed-intensity classes, pasture, and crops recover to 0.7–10.5%, together 83% of the domain's cells; area-weighted relative L^2 error is 0.20 (unweighted 0.58). The residual errors concentrate in small-area classes (forests, shrub, emergent wetland, ∼5% of cells), which now *move* — they are sensitive, unlike in the gauge experiment — but overshoot to the bounds along weakly constrained directions: ordinary semiconvergence, treatable with regularization and longer windows rather than a structural absence of signal. Three caveats keep this honest: the one-hour window samples early-transient peaks, not flood peaks (the real calibration windows are 6–12 hours and should only help); the run was capped at 15 quasi-Newton iterations; and twin marks synthesized from the truth trajectory carry no representation error, so this is an identifiability statement — an upper bound on real-data performance, not a forecast of it.

Noise turns identifiability into a signal-to-noise question. Every recovery above is noise-free, as are the verification twins behind Section 4. Such experiments establish that the gradient is right and that the observable carries the information; they do not establish that a field can be calibrated from real data. What separates the two is the size of the parameter-induced signal relative to observation error, and that ratio moves by orders of magnitude with the configuration (Table 3). On a short verification window the entire Manning-induced observation signal is a few microns of water height, so a noise-free optimizer descends on it indefinitely while a tenth of a millimetre of observation noise destroys the recovery outright — and no Tikhonov weight in a sweep up to $\beta = 1$ rescues it. At basin scale the same quantity is ∼0.1 m of peak water-surface elevation after one simulated hour, five orders of magnitude larger, which is the regime in which survey-grade high-water-mark error becomes a fair fight rather than a foregone conclusion. The lesson is not that the method is fragile: it is that friction must act long enough to imprint a signal above the noise floor, which makes window length a first-class calibration decision rather than a cost parameter. The basin-scale noisy row shows what marginal signal-to-noise looks like in practice, and why misfit at the marks is the objective but not the result. With 0.15 m of observation noise, even the *true* field cannot beat a mark MAE of about $\sigma\sqrt{2/\pi} \approx 0.12\,\mathrm{m}$ against the noisy values — and the optimizer reaches exactly that floor while driving a third of the classes onto their bounds (pasture, 428k cells, to the ceiling) and ending far from the true field it was supposed to recover. The data fit is as good as truth could achieve; the parameters that achieve it are wrong, because the last of the misfit was removed by matching noise. On real data the parameter error is invisible and the bound-pinning is its symptom — which is the concrete argument for stronger regularization, a longer window, and discrepancy-principle stopping (stop at the noise level, do not descend past it) in the production run.

Configuration	Signal	Observation noise	Outcome
Verification twin	5 μm	none	per-region recovery 4×10^{-7}
(planar, 40 steps,	water	0.1 mm	per-cell recovery destroyed ($0.20 \to 1.8$)
1.8 s)	height	1 mm	per-region destroyed ($4 \times 10^{-7} \to 0.83$, both at bounds)
30 m basin, 1 h	∼0.1 m	none	area-weighted rel. L^2 0.20; MAE $0.103 \to 0.017$ m
rain-forced, 108 marks	peak WSE	0.15 m (HWM grade)	misfit falls 9.4×, MAE $0.23 \to 0.12$ m — but 5 of 15 classes pin to bounds

Table 3: Parameter-induced observation signal versus observation error. The noise-free recoveries differ by seven orders of magnitude in the signal they descend on; only the basin-scale row is a fair test against survey-grade error (0.1–0.3 m for high-water marks). Noise is a deterministic, decomposition-independent perturbation of the twin observations, so the comparison is reproducible across rank counts.

7 The real survey: what roughness can explain

The twins establish what the machinery can recover under controlled conditions; this section turns it on the real Hurricane Harvey survey. The observable changes first — the stage gauges are unusable at this resolution — and everything after that is measured against surveyed high-water marks: the uncalibrated baseline and the drainage defect it exposes (Section 7.1), how much of the remaining error roughness can remove (Section 7.2), the calibration (Section 7.3), and the spectrum that counts how many parameters the marks can actually determine (Section 7.4). Section 8 then points the same measurement at the initial condition.

7.1 High-water marks and the uncalibrated baseline

Calibration against the actual stage records is blocked by geometry, not software: at all 13 rain-driven USGS gauges in the 30 m domain, modelled water-surface elevation sits 1–11 m *above* observed stage, because a 30 m cell stores one mean ground elevation — essentially the bank of an incised bayou — while the gauge hangs in the channel below it; the offset is time-varying as the real channel fills, so no per-gauge datum correction can repair it. We therefore changed the observable to the USGS high-water-mark archive: surveyed *peak* water-surface elevations on structures and debris lines, i.e. on the floodplain surfaces a 30 m cell does represent, and the benchmark currency of the flood-modeling literature (Inunda reports 0.67 m MAE against Harvey high-water marks [10]). Of 2,364 Harvey marks, 324 fall in the Turning domain; applying to the marks the same QC that disqualified the gauges — is the surveyed elevation above the containing cell's bed? — rejects 62% of them (they are channel surveys), leaving 108 marks of quality fair or better. The misfit compares each mark cell's peak simulated WSE over the window, $\max_k Hu(t_k)$, against the surveyed value; its adjoint needs no new machinery, because each mark's residual can be injected at its own argmax time in the segmented backward sweep of Algorithm 2, which is the exact gradient wherever the peak time is locally constant (Section 5.1). The parameter gradient of the peak misfit passes the same central-difference gates as every other block on the smooth verification twin. At basin scale the naive check fails, and a set of instrumented measurements pins down why, in a way that validates the gradient rather than the check (Table 4). No mark's argmax moves and none flips wet/dry across any probe, so the discrepancy lives below the observable; the evaluations are deterministic to nine digits and the FD value is stable across a decade of probe step, which excludes noise, curvature truncation, and sparse kinks. Decisively, the gap *accumulates with trajectory length*: at the shortest

Window	Steps	FD-vs-adjoint gap, domain-wide direction
smooth verification twin	40	9×10^{-7} (gate 10^{-5}: pass)
basin, 1 min	60	9×10^{-5} (inner-solve floor: pass)
basin, 5 min	300	8×10^{-3}
basin, 15 min	900	2×10^{-2}
basin, 30 min	1800	2×10^{-1}
basin, 1 h	3600	7×10^{-2}

Table 4: The FD gate's failure accumulates with trajectory length on identical basin, rain, marks, and converged tolerances — the signature of branch crossings along the trajectory, not of gradient error. The 30-minute and 1-hour rows sample peaks over different horizons, so strict monotonicity across them is not expected; single-class directions sit one to three orders lower at every window.

window the gate passes at the inner-solve floor with every piece of machinery engaged, and it grows three orders of magnitude by basin window lengths. A proportional defect in $\partial f/\partial n$ or the adjoint accumulation would err the same fraction at any window length; trajectory-accumulated branch crossings — the probe shifts which cells cross the depth cutoff at some step — predict exactly this profile. Two further checks close it: as the probe shrinks, the FD converges to the adjoint direction by direction — the largest single-class direction passes the 10^{-5} gate outright at the smallest probe — and the fifteen per-class gaps *sum* to the domain-wide gap, the additivity that independent per-cell branch crossings predict and no plausible defect would arrange. The measured conclusion: the adjoint returns the exact one-branch derivative; over long windows the objective is dense with wet/dry branch surfaces, and a finite-step central difference converges to a locally *smoothed* slope that sits several percent from any one branch. The quasi-Newton iteration consumes one-branch derivatives of a piecewise-smooth objective — standard practice for max-type and switching systems — and its observed descent (Section 6.1) is consistent with that footing (Section 5.1).

The remaining scientific choice — confirmed or corrected by the domain team — is the calibration window, which for marks that record the event maximum must cover the flood peak.

The uncalibrated baseline against the real survey. Because the marks record the event maximum, the honest baseline evaluation must simulate through the crest: a 72-hour rain-forced forward from the available initial state (2017-08-26 18:00, reaching past the Buffalo Bayou crest), NLCD-prior Manning, free outflow, backward Euler at $\Delta t = 1\,\mathrm{s}$ on the 2.93M-cell mesh — 259,200 implicit steps, *zero* Newton failures, two hours on one GPU node. The model wets all 108 marks, and the uncalibrated peak-WSE error decomposes into two distinct populations (Table 5; confirmed mark for mark at fully converged inner tolerances). The marks that crest inside the window carry the error a Manning field can plausibly address. The others never crest at all: their cells fill monotonically through the entire window — every one peaks at the final hour with zero recession, still rising while the first population is already draining — accumulating some ten metres of standing water whose late-window inflow arrives laterally, after the rain has ended. Those cells sit in the downstream Buffalo Bayou reach: the model cannot drain it, a conveyance/outlet defect that no roughness value can absorb and that would corrupt any calibration asked to fit it. The obvious suspicion — that we simply failed to give the domain an exit — does not survive inspection of the mesh. The domain is a delineated watershed rather than a rectangular cutout (35 of its 6,198 boundary nodes lie on the bounding box), so its unflagged perimeter is a catchment divide and a reflecting wall is the right condition there; and its single free-outflow side set sits at the perimeter's lowest point, the only boundary nodes at $z \approx 0$ against a boundary median of 29.9 m. The exit is

Marks	n	MAE	bias
crest inside the 72-h window	71	1.51 m	+1.20 m
never crest (reach cannot drain)	37	7.06 m	+7.06 m
all	108	3.41 m	+3.21 m

Table 5: The uncalibrated baseline against the 108 surveyed high-water marks, decomposed. The second population is model-high at every single mark and identifies a reach with no working exit; the first is the calibration target.

Group	n	mean lon.	bias	MAE
crest h29–41 (upstream)	46	−95.695	+0.30 m	0.72 m
crest h48–72 (middle)	25	−95.568	+2.85 m	2.96 m
never crest (downstream)	37	−95.440	+7.06 m	7.06 m

Table 6: The crest-timing groups are spatial bands, and the model-high bias grows monotonically downstream toward the reach with no working exit. The upstream band is the calibration target; the middle band is contaminated by the same defect at intermediate strength and cannot serve as held-out data.

present and correctly placed. What the ponded reach then reveals is a limitation of the idealization itself: it holds water above the elevation of parts of its own divide, which a reflecting wall retains and the real event would have spilled. The calibration target is therefore the 71 admissible marks; the drainage defect is reported as a finding in its own right; and the remaining bundle (initial-condition, rainfall, mesh-elevation, and datum error) is what the calibration must be prevented from absorbing along with the roughness signal. The literature's calibrated benchmark on this event is 0.67 m MAE [10].

The defect is a gradient, not a switch. Sorting the marks by crest time splits them into three groups, and those groups turn out to be three *spatial bands* whose bias increases monotonically downstream (Table 6). The never-crest population is the end of a continuum, not a separate pathology: the intermediate band already carries +2.85 m of the same model-high signature. Two consequences follow. First, a calibration window has to be placed on the upstream band, where the model is physically sound — crests there fall in event hours 29–41, and the 46 marks in that band are the target of every production number below. Second, the intermediate band is *not* clean held-out data, which rules out the obvious validation design; out-of-sample skill has to be tested by cross-validation within the upstream band instead.

7.2 How much of the error can roughness explain?

Before asking an optimizer to fit fifteen land-cover classes it is worth asking what any roughness field could achieve, and that question has an optimizer-independent answer. Scaling the entire NLCD field by a single factor α traverses the natural first direction of the problem — close to, though as Section 7.4 measures, not exactly, its most identifiable one; evaluating the peak-WSE misfit along it bounds what we will call the *authority* of roughness over this observable — the amount of the model–survey error that changing roughness can remove — without reference to how, or whether, a calibration converges.

We ran that scan on the production configuration: a 12-hour window over the upstream crest cluster (event hours 29–41), initialized from an hourly checkpoint of the 72-hour forward and with

α	n range	J	MAE	dry
0.20	0.005–0.032	implicit solve diverges		
0.30	0.008–0.048	6.737×10^2	0.6392	1/46
0.45	0.012–0.072	7.038×10^2	0.6614	1/46
0.60	0.016–0.096	7.406×10^2	0.6776	1/46
0.70	0.019–0.112	7.655×10^2	0.6894	1/46
0.80	0.022–0.128	7.813×10^2	0.6991	0/46
1.00 (NLCD)	0.027–0.160	8.083×10^2	0.7188	0/46

Table 7: Uniform roughness scan on the production configuration (12-hour window, 46 cluster-A marks); every cell carries $n = \alpha\, n_{\mathrm{NLCD}}$, so the n range column is the whole lookup table scaled by α. J falls monotonically toward lower roughness with no interior minimum: the entire uniform-roughness knob is worth 0.08 m against a 0.72 m error, and only at roughness 30% of published values, where n reaches 0.008 and smooth concrete is ≈ 0.012. Within a defensible range it is worth about 3 cm ($\alpha = 0.70$, itself the optimum implied by the prior width of Section 4). The limit at $\alpha = 0.2$ is numerical, not drying — one mark stays dry throughout.

the rain re-aligned to the window start (both validated bit-exactly), scored against the 46 surveyed marks whose modelled crest falls inside the window. Each point is one forward with no adjoint (Table 7).

The window is set by the observable. A high-water mark records an event maximum, so it carries information about roughness only if its crest occurs inside the calibration window; before its crest it is an inequality, and after it a constant. The measured crest times decide the window rather than any budget: they are bimodal, clustering in event hours 29–42 and 60–72 with an empty gap between, so a window spanning both would spend most of its length integrating marks that had already peaked. The upstream cluster is also the population the model can legitimately be scored against at all, since the downstream marks sit in a reach it cannot drain (Section 7.1) — a defect no window length repairs. The 46 marks used throughout this paper are therefore those for which the observable means what it is meant to mean, and every number reported here — the uncalibrated baseline, the scans, and the calibration — is measured on that same window, so they compose. The one exception is the spectrum of Section 7.4, measured against the same 46 marks but on a 7,200-step (two-event-hour) pilot of this configuration rather than the full 43,200-step window; Section 7.4 states the gap, and re-measuring it on the production window is planned before submission.

The entire uniform-roughness knob is worth 0.08 m against a 0.72 m error (Figure 2) — about 11% — and it reaches that only at $\alpha = 0.3$, i.e. Manning values 30% of the published NLCD lookup, as low as $n = 0.008$ where smooth concrete is ≈ 0.012. Over a physically defensible range ($\alpha \gtrsim 0.7$) the knob is worth about three centimetres. The curve is mildly concave with no interior minimum: J falls monotonically toward lower roughness until the implicit solve fails.

That number is corroborated from an independent direction. The identifiability twin of Section 6.1 measured the peak-WSE signal induced by a large class perturbation at ~ 0.1 m — the same authority scale, from a different experiment. So 85–90% of the model-versus-survey residual is not roughness at all: it is rainfall forcing, mesh elevation, datum, and representation error at 30 m.

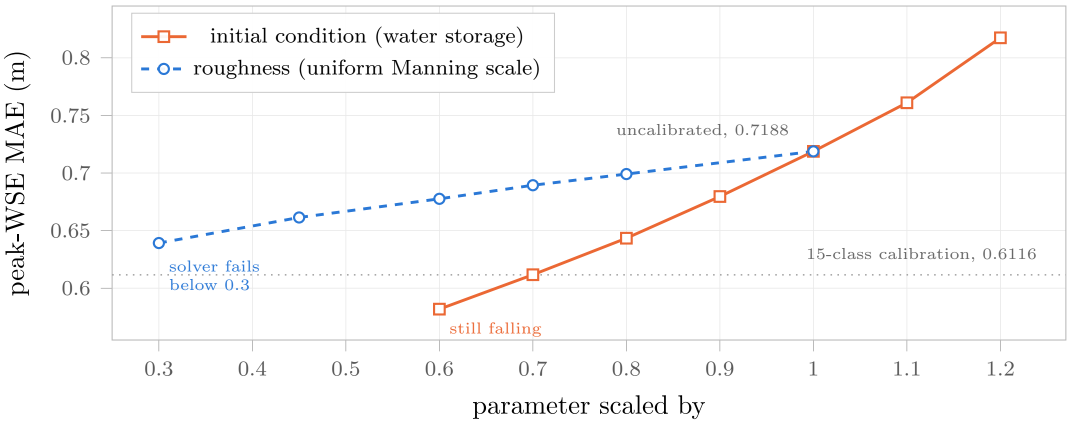

Figure 2: What each parameter can do to the peak-WSE error, on identical axes: the horizontal axis scales the parameter, the vertical axis is the same mean absolute error over the same 46 marks, and both curves pass through the uncalibrated model at 1.0. Roughness is concave and runs out — and its useful end is physically indefensible, reaching $n = 0.008$ before the implicit solve fails. The initial condition is about four times steeper per unit fractional change, convex, and has not turned over at a 40% reduction in antecedent water. The dotted line is the fifteen-class calibration, which lies off both curves because it redistributes roughness rather than scaling it.

Field	J	ΔJ	MAE	ΔMAE
NLCD prior	808.3	—	0.7188	—
developed classes 21–24 only (71.6% of cells)	786.8	−21.5	0.7306	+0.0118
everything else (28.4%)	827.5	+19.2	0.7268	+0.0080
both (= uniform $\alpha = 0.45$)	703.8	−104.5	0.6614	−0.0574

Table 8: Each half of the domain scaled alone makes the fit *worse*; scaled together they help by 45× the sum of the parts. Peaks are set by path-integrated conveyance along flow paths that cross both groups, so only domain-coherent changes drain them.

The observable constrains one mode, not fifteen. A natural objection is that spatial redistribution should beat uniform scaling — that fifteen classes have freedom a single α does not. We tested it directly by scaling only part of the domain to $\alpha = 0.45$ and leaving the rest at the NLCD prior (Table 8).

Each half alone makes the fit *worse*, and the two together help by 45× the sum of the parts — a strong positive interaction, not two competing signs. The mechanism is consistent with every other measurement: a mark's peak is set by the *path-integrated* conveyance from upstream to that mark, and those paths cross both groups. Speeding part of a path does not drain it; the resulting fast/slow junction ponds water, so partial changes *raise* peaks. Only a domain-coherent change lowers them.

The conclusion is a design finding, not a tuning observation. The peak-WSE observable at these marks constrains essentially *one* roughness degree of freedom — the domain-scale, path-integrated conveyance — worth $\sim 10\%$ of the model–survey discrepancy. It does not carry fifteen classes' worth of information. This unifies the earlier evidence rather than sitting beside it: the perfect-data twin recovers only the classes that move the domain mean, and the noisy twin pins the others to bounds precisely because they carry no signal to fit, so the optimizer fits noise in those directions instead.

Manning field	MAE	Δ	physically defensible?
NLCD lookup (uncalibrated)	0.7188	—	yes, by construction
uniform $\alpha = 0.70$	0.6894	-0.029	yes
uniform $\alpha = 0.30$	0.6392	-0.080	no (n to 0.008)
calibrated, 15 classes	**0.6116**	$\mathbf{-0.107}$	mostly; one class at a bound

Table 9: Peak-WSE MAE against the 46 cluster-A marks. The calibration beats the entire uniform-roughness knob — including its physically indefensible end — and beats the best *defensible* uniform field by 0.078 m. Spatial redistribution is therefore worth roughly three and a half times what a single global scale can reach.

Relation to equifinality. None of this is a new phenomenon: that many different parameter sets reproduce a hydrological observation about equally well is Beven's equifinality thesis [22, 23], and the usual response is to characterize the acceptable set by sampling it. What an adjoint changes is the cost of the diagnosis. Where GLUE-style methods *sample* the equifinal set with many forward runs, the measurements above *measure* its dominant direction directly: four forward evaluations bound the authority of the whole roughness field, and two more show that the modes are not separable. The result is the same kind of statement, obtained at a cost that does not grow with the number of parameters — which is what makes it affordable before a calibration rather than after one.

Why this generalizes. The measurement uses nothing specific to roughness. Given a parameter, an observable, and a model error to explain, the same three steps — scan the dominant parameter direction, check whether sub-domain changes are separable, and compare the resulting authority against the residual — bound what calibrating that parameter can ever accomplish, before any optimizer is written. We suggest it as a routine precondition for calibration studies, which commonly report the misfit reduction achieved without establishing the ceiling it was approaching.

7.3 The calibration against the real survey

The scan sets the bar a multi-parameter calibration must clear to justify its parameters. Running the class calibration of Section 4 on the same configuration — fifteen NLCD classes, relative variables, prior width as discussed below, starting from the NLCD lookup — clears it (Table 9).

Two things about this number deserve emphasis, one supporting and one limiting.

Supporting: the comparison that matters is against $\alpha = 0.70$, not $\alpha = 0.30$. The latter is the misfit's unconstrained preference and reaches $n = 0.008$, which no one would defend; the former is the optimum implied by the same prior the calibration itself uses, and is the honest one-parameter competitor. Against it the calibration wins by 0.078 m, so the fifteen parameters are not decoration — *letting roughness vary by class removes error that no global scaling can*, which is a stronger statement than the scan alone justifies and the one the half-domain experiment (Table 8) would not have predicted.

Limiting, and there are three. First, the objective and the MAE do not move together. The calibrated field's misfit is $J = 615.4$; sliding along the uniform scan's measured MAE-versus-J relation to that same objective gives ≈ 0.596 m, where the calibrated field scores 0.6116, so redistribution buys objective slightly less MAE-efficiently than scaling does and the two should not be quoted interchangeably. That comparison is a counterfactual, though, and in the calibration's favour: $J = 615.4$ lies below 673.7, the lowest objective the uniform knob reaches before the implicit solve fails, so the uniform field being compared against cannot actually be run. *The calibration* *reaches an objective global scaling cannot attain at all.* Second, the run is nine BLMVM iterations from the NLCD prior and was stopped by the 12-hour wall rather than by a tolerance: J fell 18.7% overall but only 0.01% on the last iteration while the gradient norm fell 8.5×, which is convergence behaviour without being a converged run, and developed-medium sits at its lower bound throughout; Section 4 discusses why that bound is partly an artifact of how the prior was stated. Third and most important, **0.6116 m is an in-sample score**: the 46 marks that define the objective are the 46 marks it is evaluated on. We report it as a fit, not as skill. Out-of-sample validation has to be cross-validation *within* the upstream band, because the middle band carries the drainage defect (Table 6) and would not test generalization so much as re-measure that defect.

Section 7.4 makes the second and third caveats concrete rather than conventional: the calibrated field lies 9.2 prior standard deviations from the NLCD lookup, and only 2.9% of that displacement is in a direction these marks constrain.

Set against the error budget: the model–survey discrepancy on these marks is 0.72 m and roughness, calibrated as well as we can currently calibrate it, accounts for 15% of it. The remaining 85% is rainfall forcing, mesh elevation, datum, and representation error at 30 m — and Section 8 shows that one of those is measurable with the same method, and larger.

7.4 How many parameters does the observable support?

Everything above infers the answer from scans. The eigenvalues of the prior-preconditioned Hessian give it directly. In coordinates where the prior is white, the posterior precision is $I + \Gamma^{1/2} H_{\mathrm{mis}} \Gamma^{1/2}$, so each eigenvalue λ_i of the whitened misfit Hessian compares data information against prior information in one direction: the count with $\lambda_i > 1$ is the number of parameters the observable supports, and the eigenvectors say which *combinations* of classes those are — which no scan can.

Assembling it. With fifteen classes this needs no low-rank machinery, but it does need the right matrix. Differencing the *gradient* field fails here, and instructively: over a window of this length the objective is dense with wet/dry branch surfaces (Section 7.1), and while the adjoint is exact on its own branch the gradient jumps between branches, so a difference quotient samples a discontinuity. Our first attempt returned a matrix with 45% relative asymmetry and eigenvalues to −6.8, which its own consistency check rejected. We therefore assemble the *Gauss–Newton* Hessian from observation sensitivities,

$$H = \tfrac{1}{\sigma^2} S^\top W S, \qquad S_{ik} = \partial(\max_k Hu)_i / \partial \alpha_k, \tag{6}$$

which differences the mark peaks rather than the gradient. This works for the reason Section 5.1 established — mark argmaxes do not move under small probes — and we verify it rather than assume it: across all 690 mark–column pairs only 27 (3.9%) changed their peak time. It costs 16 forwards rather than 16 forward-plus-adjoints, and it is positive semi-definite by construction, so the failure mode that invalidated the first attempt cannot recur silently. One caveat of scope: the spectrum is measured on a 7,200-step (two-event-hour) pilot of the production configuration, against the same 46 marks, while the calibration it interprets runs on the full 43,200-step window. A longer window should raise the sensitivity each peak integrates, and with it λ; re-measuring the spectrum on the production window costs sixteen forwards and is planned before submission.

The answer is one. $\lambda = 3.03,\ 0.64,\ 0.33,\ 0.32,\ 0.21,\ 0.10, \ldots$ (Figure 3) — a single eigenvalue above unity, a spectral gap of 4.8, and 2.02 degrees of freedom for signal in Rodgers' sense. **These 46 marks support one roughness parameter** — one combination of the fifteen class values on

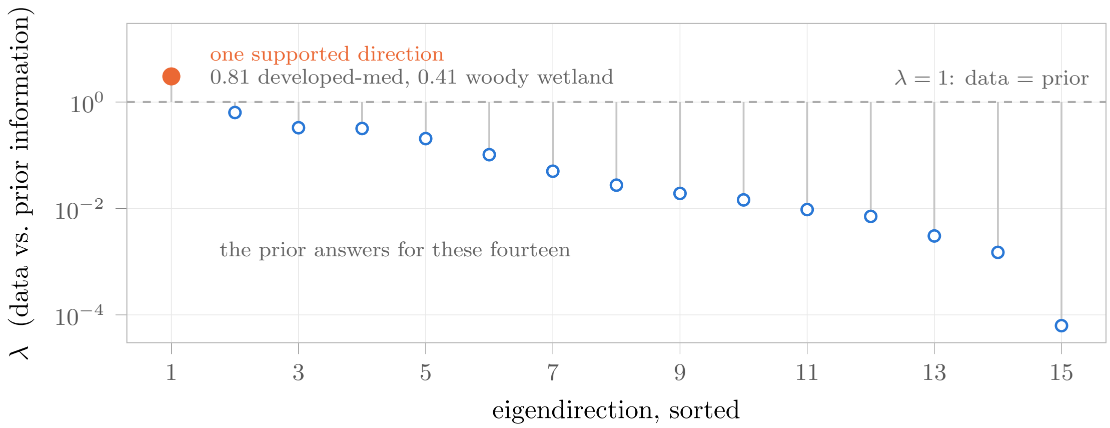

Figure 3: Eigenvalues of the prior-preconditioned Gauss–Newton Hessian over the fifteen land-cover classes, sorted. Above $\lambda = 1$ the marks say more than the prior does; below it the prior answers. One direction clears the line, and the remaining fourteen fall away over four decades — so a fifteen-parameter calibration is fitting one parameter and inheriting fourteen from its prior. The log axis is not cosmetic: on a linear one the cliff, which is the result, would be invisible.

which the data out-weighs the prior, and no second. The scans and the spectrum agree, and they are independent: no optimizer, no line search, and no calibration enters Eq. (6).

But the scan varied nearly the wrong direction. The leading mode, with components -0.81 on developed-medium, $+0.41$ on woody wetland, -0.36 on developed-low and -0.18 on developed-high, is dominated by the large-area classes. Its overlap with the uniform direction the α-scan varies — equal weight per *class* — is 0.263, which for fifteen classes is indistinguishable from random (0.258). Its overlap with the same physical change weighted per *cell* is 0.768. So the informative mode is the domain-scale conveyance mode we claimed, but area-weighted; the α-scan probes it only obliquely. That is why the class calibration beats uniform scaling by 0.078 m (Table 9): it is not that fifteen parameters beat one, it is that the scan's one parameter was the wrong one.

What the calibration actually did. Projecting the calibrated displacement onto the eigenbasis is worth reporting, and it cuts both ways. The calibrated field sits **9.2 prior standard deviations** from the NLCD lookup, and of that motion only 2.9% lies in the single data-constrained direction; 85.6% lies in directions where data and prior are comparable and 11.5% where the prior alone determines the answer. Per class (Table 10) the marks shrink the prior uncertainty by 36% on developed-medium and by 12–17% on four more, and by 5% or less on the remaining ten.

The reassuring half is that the null space behaved. Two iterations in, the optimizer had moved deciduous forest to $\alpha = 0.739$, mixed forest to 1.096 and cropland to 1.274 — three classes these marks teach nothing about — and by iteration nine accumulated curvature had returned all three essentially to the prior (0.983, 1.006, 1.098). Those excursions were transients of an under-informed quasi-Newton approximation rather than features of the minimizer: the misfit gradient in uninformative directions is not zero, merely uninformative, and given enough iterations the prior wins. A calibration stopped early would have reported them as results, which is an argument for running these problems to convergence even when the objective has visibly flattened.

The unreassuring half is what survives. The residual 9.2σ is concentrated in the two *best*-informed classes, and both are pressed against the edge of what the prior permits: developed-

NLCD	class	prior σ_n	posterior σ_n	learned
23	developed medium	0.015	0.010	36%
22	developed low	0.015	0.012	17%
90	woody wetland	0.015	0.012	17%
81	pasture	0.015	0.013	15%
21	developed open	0.015	0.013	12%
24, 95, 11		0.015	0.014–0.015	2–5%
41, 43, 82, 31, 42, 71, 52		0.015	0.015	$\leq 1\%$

Table 10: What the 46 marks teach about each land-cover class, as the fractional reduction from prior to posterior standard deviation. Seven classes learn essentially nothing: any value a calibration reports for them is the prior speaking, not the data.

medium sits at its $\alpha = 0.3$ lower bound for all nine iterations, and woody wetland is driven to $\alpha = 0.355$, some 4.2σ from its lookup value. So the displacement is not an optimizer wandering where the data is silent. It is the data asking, in the one direction it can speak, for roughness below what the published table and our stated prior width jointly allow. Whether that request is admissible is not something the optimizer can settle; it is the first of the questions we put to the domain team.

A direct test: calibrate three classes instead of fifteen. Everything above is a statement about a matrix. It makes a prediction an optimizer can falsify: if the marks really inform one direction, then calibrating only the classes that carry that direction should recover most of what all fifteen achieved — and if it does not, the information lives outside the leading eigenspace and the one-parameter reading is too narrow. We ran it. Taking the three classes with the largest components in the leading eigenvector (developed-medium, woody wetland, developed-low), freezing the other twelve at the prior, and calibrating from the NLCD lookup gives $J = 682.1$ after four iterations — the last two changing J by 0.02% and 0.001%. Against the NLCD prior's 808.3, three parameters remove 126.1 units where fifteen remove 150.9: 84% **of the reduction from** 20% **of the parameters.** The twelve frozen classes were verified to sit bit-exactly at the prior in the output, so the comparison is what it claims to be.

Two details sharpen it. The remaining 16% is real, not noise, and it decomposes exactly: the fifteen-parameter field pays 8.8 more units of prior penalty to buy 33.5 more units of misfit reduction. So the twelve non-leading classes are not literally decoration — in aggregate they reach a few per cent of J — but they are not where the information is. And the three parameters land where the fifteen-parameter calibration put them: developed-low at $\alpha = 1.626$ against 1.627, developed-medium on its $\alpha = 0.3$ bound in both. Two classes reproduced to three digits from a different starting point and a different active set is the behaviour of parameters the data actually determines. Woody wetland is the exception and is instructive: alone it dives to $\alpha = 0.301$ where the full calibration left it at 0.355, because with its partners frozen it must supply their share of a change the marks constrain only in combination — equifinality, exactly where the eigenvector says to expect it.

Would more marks fix it? This is the natural objection — 46 observations for 15 parameters sounds like too few — and Eq. (6) answers it, because λ scales linearly in the number of comparable observations and as σ^{-2} in their quality (Table 11). The answer is that more marks help, slowly, and that the supply is exhausted long before the parameterization is justified. Going from the

Marks	Supported	That data set is
46	1	the cluster-A marks (this study)
71	1	all admissible marks, 24-hour window
108	2	every QC-passed mark, 72-hour window
324	5	every Harvey mark in-domain, *before* QC
7×10^5	15	—

Table 11: How the supported-parameter count scales with survey size at fixed quality ($\sigma = 0.15\,\mathrm{m}$); a parameter is *supported* when the data narrows it more than the prior does ($\lambda > 1$). At fixed size, improving quality to $\sigma = 0.10\,\mathrm{m}$ buys a second parameter and $\sigma = 0.05\,\mathrm{m}$ — better than any high-water-mark survey achieves — buys five. The scaling holds the measured per-mark sensitivity fixed, so the longer-window rows describe mark populations only; a longer window also raises each mark's sensitivity, which this table does not credit.

46 marks of this window to *all* 71 the model can legitimately be scored against still supports one parameter. Every QC-passed mark in the domain, over a window long enough to catch all of them, supports two. Reaching five would take every Harvey mark that falls in the domain including the 62% that failed QC because they are channel surveys a 30 m cell cannot represent. Fifteen would take of order 10^5 marks.

The conclusion is therefore not that Manning roughness is unidentifiable. It is that *surveyed peak water-surface elevations at survey accuracy carry about two degrees of freedom of information about distributed roughness in this basin*; that this can be computed from sixteen forward runs before any optimizer is written; and that the limitation is a property of the observing system rather than of the method. A study that reported only its misfit reduction would have concluded the opposite.

8 Where else the error lives: the initial condition

Roughness, calibrated as well as we can calibrate it, explains 15% of this model's error against the marks. Something explains the other 85%, and the candidates — rainfall forcing, mesh elevation, datum, representation error, the state the window starts from — are not all equally measurable. This section points the ceiling measurement of Section 7.2 at the one candidate it can probe with a single scan direction: the initial condition. The initial state is also the input a variational initialization would control, so the measurement doubles as a feasibility probe for initial-state estimation with the same adjoint machinery, which the conclusions take up.

Our window begins at event hour 29 from a checkpoint of the 72-hour forward, and the natural single direction to scan is the amount of water that checkpoint holds. Scaling all three conserved variables by a common factor a changes antecedent storage while leaving every velocity unchanged, since $u = (hu)/h$ is invariant. Each point is one forward (Table 12).

Three things follow, and they matter more than the roughness result.

The initial state has about four times the authority of roughness (Figure 2). Per unit fractional change, $\mathrm{d}\,\mathrm{MAE}$ is 0.392 for the initial condition over its well-sampled range ($a \in [0.8, 1.1]$) against 0.098 for a uniform roughness scale. Like for like at a 20% perturbation: 0.075 m versus 0.020 m. And the plausibility runs the same way — a 20% error in water stored after a 29-hour spin-up under radar-derived rainfall is far easier to defend than Manning values at 30% of the published table.

The response has the opposite curvature to roughness's. $J(\alpha)$ for roughness is concave

IC scale a	J	MAE	ΔMAE per 0.1
0.80	6.652×10^2	0.6434	—
0.90	7.284×10^2	0.6796	+0.0362
1.00 (checkpoint)	8.083×10^2	0.7188	+0.0392
1.10	9.071×10^2	0.7610	+0.0422
1.20	1.025×10^3	0.8174	+0.0564

Table 12: Peak-WSE misfit against antecedent water storage. The response is monotone over the whole range with no interior optimum, straight to about 1% between $a = 0.8$ and 1.1 and convex above it: adding water hurts at an accelerating rate. The model holds systematically too much water at hour 29. No mark is dry at any point.

and saturating — the knob runs out — consistent with a parameter acting indirectly through path-integrated conveyance. $J(a)$ for the initial state is straight to about 1% between $a = 0.8$ and 1.1 and *convex* above, steepening as water is added, and it does not turn over anywhere in the range. Extending the scan below the table confirms it: at $a = 0.6$ — a 40% reduction in antecedent water — the MAE is still falling, to 0.5818 m (Figure 2). Neither the slope nor the sign gives any indication of an optimum near the checkpoint value. That is the signature of a systematic bias rather than a poorly-tuned parameter: the model does not hold slightly the wrong *distribution* of water at hour 29, it holds too much of it.

It is a diagnosis, not a knob. We are not proposing to calibrate the initial condition against these marks. Over this window the initial state carries 3 unknowns on each of 2.93M cells against 46 peak observations, and everything Section 7.2 establishes about identifiability applies with three more orders of magnitude of freedom: such a calibration would fit the marks essentially exactly and mean nothing. What the scan does establish is the weaker and more useful claim that the residual these marks measure lives substantially in the water balance — in the rainfall forcing or the antecedent state — rather than in the friction, which is where a roughness calibration alone would have kept looking.

One caveat bounds all three. The checkpoint is a *model* state, not an observation, so it already contains accumulated forcing and elevation error; the scan perturbs that state rather than an independent measurement of it. A calibration of the initial condition would therefore absorb such error rather than correct it, and could quietly launder the downstream drainage defect of Section 7.1 into a "corrected" initial state — the same trap as scoring against the 37 marks the model cannot drain. Establishing whether a 20% water-balance error at hour 29 is physically plausible is a question for the forcing data, not for the optimizer, and it heads our list of questions at the end of this draft.

9 Conclusions and outlook

An exact, verified Jacobian turns out to be the single artifact from which everything else follows cheaply: adjoint state and parameter sensitivities, gradient-based calibration, and implicit stepping. The capability is production-ready in the sense that matters for an operational code — behind a flag, FD-gated in CI, and with the verification baseline (FD coloring) kept alive in the configuration system.

The second artifact is a measurement rather than a capability, and it is the one we would most like others to copy. Asking what fraction of a model's error a parameter is *able* to explain is cheap, optimizer-independent, and it reframes what a calibration result means. Here it bounded roughness at 4% of the error within defensible values and 11% at the edge of plausibility; the calibration then reached 15%, beating every global scaling and confirming that letting roughness vary by class removes error a single global factor cannot. What makes the measurement worth copying is not that number but the two it led to. Pointed at the residual instead of the parameter, the same machinery isolated a reach the model cannot drain; pointed at the initial condition, it found more authority than roughness has. A calibration study that had only reported its misfit reduction would have found neither.

Next steps, in order. First, cross-validate the calibration reported here — nine quasi-Newton iterations stopped by a wall-clock limit, with one class on a bound, is a fit, not demonstrated skill — and the held-out marks must come from within the upstream band, because the middle band carries the drainage defect. Second, calibrate the number of parameters the observable supports rather than the number the land-cover map offers; the spectrum of the prior-preconditioned Hessian would settle that count directly, and at fifteen parameters it is a small dense eigenproblem rather than a research project. Third, the water balance: the initial-condition scan says the antecedent state carries more of this residual than the friction does, which makes the forcing and the initial condition — not the roughness — the next thing to calibrate or diagnose, with the machinery applying unchanged and dJ/du_0 already verified. Fourth, the conveyance and outlet representation the drainage finding implicates. Finally, coupling the gradient path to PyTorch through the existing PETSc–PyTorch bridge for learned-physics and hybrid data-assimilation experiments.

A closing note on cost. Checkpointing makes the long-window adjoint memory-feasible, but at the 30 m mesh's CFL-limited $\Delta t = 0.25$ s a full 36-hour Hurricane Harvey gradient still costs several forward sweeps over half a million steps. The affordable path stacks reductions that are all now in place. First, fully implicit stepping at CFL-free time steps: backward Euler with the assembled Jacobian carries the 30 m, 2.93M-cell Harvey-region mesh at $\Delta t = 1$ s — a step the differentiable explicit path cannot take at any tested size (Section 5) — with Newton converging in about two iterations per step behind a block-Jacobi-preconditioned Krylov solve at converged inner tolerances; larger implicit steps are under investigation. Second, GPU execution of the entire gradient pipeline: the right-hand side, the assembled Jacobian and its transpose solves, the preconditioner setup, the parameter Jacobian, the observation operator, and the revolve-checkpointed trajectory are all device-resident. On four A100 GPUs, a forward-plus-adjoint gradient of a one-simulated-hour window on that mesh (3,600 implicit steps, 400 in-memory checkpoints) takes about nine minutes, and one TAO iteration of a dense-gauge twin calibration (418,076 gauges, 39-step window) takes 1.7 s — roughly 19× one 64-core CPU node of the same system. Device and host runs produce bitwise-identical trajectories under a common compiler and gradients identical to every printed digit, so the finite-difference gates of Table 1 carry over unchanged. Third, land-cover-class parameterization (tens of TAO evaluations instead of hundreds) and calibration windows placed on the storm peak rather than the full event.

Red/tiger-team review: the weakest parts of this draft

[Temporary section, removed before submission. Our own adversarial read of the draft — the places a hostile reviewer (or a careful friend) will push, ranked by how much damage they could do, each stated as of today. Cross-references Qn point into the numbered issues of the next section.]

1. **The calibration is real but not converged.** The abstract's differentiator — production code made differentiable, versus reduced solvers rewritten in ML frameworks — invites comparison with Inunda's calibrated 0.67 m MAE against surveyed Harvey high-water marks, and that comparison must not be made: the mark sets differ, and ours is a favourable upstream

subset chosen because the model drains it. What now exists is a calibrated field that beats every uniform alternative (Table 9), reached from the NLCD prior by the scaled optimization problem of Section 4. The first attempt took *zero* steps — a scaling defect in the optimization problem, not in the gradient, diagnosed, fixed and gated — and the exposure that leaves is worth stating plainly: a reviewer may reasonably ask whether the negative identifiability result is a property of the problem or of an optimizer we could not drive. The α-scan answers that independently of any optimizer, and the calibration now lands where the scan implies it should. What is still missing is convergence: the reported field is two quasi-Newton iterations in, one class rests on a bound, and there is no out-of-sample test — which must be cross-validation *within* the upstream band, since the middle band carries the drainage defect (Table 6). (Q1, Q2)

2. **The identifiability caveats are weaker than they were, but the twins still carry them.** The measured study (Section 6.1) — 13 gauges fail, 108 marks identify the classes covering 83% of the domain — rests on a one-hour window, a 15-iteration cap and no representation error, which is exactly where a reviewer will push. Section 7.2 now answers that push from a different direction: the α and half-domain scans are on the *production* 12-hour window against *real* survey data, carry no iteration cap because no optimizer is involved, and reach the same conclusion the twins did. The remaining exposure is narrower — the twins are still the only evidence about *which* classes are recoverable, as opposed to how many. (Q2)

3. **Long-window FD gates measure the objective's nonsmoothness, not gradient error — and the paper must not be read as claiming more.** The gradient is verified to the inner-solve floor on short windows at the production tolerances (which are now converged tolerances, closing the old verified-versus-production gap), and the long-window failure of the naive gate is measured and explained (Section 7.1). The exposure is a reviewer reading "FD-gated in CI" as covering production window lengths. (Q3)

4. **"Implicit stepping is required" means required for the differentiable right-hand side.** The operator-split friction the forward-only code uses is stable under explicit stepping but buries Δt inside the RHS; the differentiable RHS is friction-stiff ($\Delta t \lesssim 2.4\,\mathrm{ms}$ on the spun-up 30 m state) and diverged at every tested explicit step. Any wall-clock comparison against operator-split explicit stepping compares different discretizations and is labeled as such. Still open: $\Delta t \geq 5\,\mathrm{s}$ is blocked by a preconditioner failure. (Q6)

5. **We changed the physics to fix the solver, and the changes await domain sign-off.** The regularized drag velocity ($h_{\mathrm{anuga}} = 0.001\,\mathrm{m}$, chosen as "smallest that cures the stall"; 0.01 m observed unstable and the mechanism only conjectured) and the transmissive outlet (replacing critical outflow; alters outlet physics; weakly reflective for subcritical flow) are both defensible and both flagged opt-in, but a reviewer can fairly ask whether the calibrated n compensates for boundary and regularization choices rather than physics. Sensitivity of the calibrated field to h_{anuga} and to the outlet choice should be demonstrated, not asserted. (Q4, Q5)

6. **Noise response is measured but only partly built into the method.** The basin-scale noisy twin shows the failure mode (misfit falls 9.4× while five classes pin to bounds, Table 3). The Tikhonov weight is no longer chosen by judgment — it is σ_n^{-2} for a stated prior standard deviation on Manning n, Eq. (5) — but that only relocates the judgment into σ_n, which the domain team has not yet fixed, and the value we are using (0.015) is close to the largest that leaves the dominant direction with an interior minimum. Discrepancy-principle stopping still

does not exist, and in any case the discrepancy target $J \approx N/2$ is unreachable here: the achievable floor is set by structural error an order of magnitude larger than the 0.15 m survey error, so the floor must be reported as measured rather than assumed. (Q7)

7. **The peak-WSE argmax remains formally open.** Its kinks are measured absent at this geometry twice over — no argmax moved across any FD probe, and the production objective is bit-identical across three domain decompositions — but a remedy is proposed (frozen-argmax outer iteration) and not implemented, and the theoretical footing is the standard one for max-type observables rather than a theorem. (Q2, Q3)

8. **Reproducibility currently rests on an unmerged PETSc fork.** The device-resident assembled-Jacobian path (blocked-COO BAIJ assembly on GPU) exists only on our PETSc branch. Until it is upstreamed or the paper's results are reproducible with stock PETSc (AIJ path, at some cost), the artifact story is weaker than the "behind a flag, FD-gated in CI" framing suggests. (Q8)

9. **The novelty claim is narrowed twice over, and what is left is not the adjoint.** "First E3SM component with built-in discrete-adjoint parameter estimation" was wrong (MALI inverts for basal friction with an AD adjoint of its steady velocity solver [18, 17]); the claim now reads first *transient* discrete adjoint in an E3SM component. A reviewer who knows MITgcm will press further and should: an exact AD adjoint of a production ocean model has driven ECCO estimation for two decades [15, 16], so "production code made differentiable" is a route with a long pedigree, not an innovation. Nor is the observable ours: Pujol et al. assimilate high-water marks variationally for distributed friction, and use adjoint sensitivity to choose the parameterization [9]. What survives is (i) the exact assembled Jacobian serving the implicit solve and the adjoint from one artifact, with the argument that implicit stepping is required for a differentiable friction RHS, (ii) basin scale on GPU, and (iii) the authority measurement of Section 7.2, which we have not seen reported for any flood-model parameter. The draft should lead with (iii). Remaining check before submission: ocean/MPAS adjoint efforts, and whether any E3SM land-model calibration uses a true adjoint rather than ensemble or surrogate gradients.

Questions we need domain guidance on

[Temporary section, removed before submission. Five questions where a domain judgment, not another run, is what is missing. Each states what we measured, what we need, and what changes depending on the answer. The longer list of open issues, tagged by team, follows.]

1. **How far can the NLCD Manning lookup be wrong, in percent?** *This one sets a number in the method directly.* The prior width is the regularization: $\beta = \sigma^{-2}$. We measured that holding the calibration to a physically defensible field requires believing the lookup is good to $\pm 15\%$; allow $\pm 30\%$ and the data drives developed-medium urban to $n = 0.036$ (smoother than concrete) while developed-low, interleaved with it, goes to 0.24. We are currently using a width at the tight end of that range and saying so. *If* $\pm 15\%$ *is defensible*, the calibration stands as reported. *If the honest width is* $\pm 30\%$ *or more*, then the headline result is that this observable cannot support a defensible distributed roughness field at all — a stronger negative claim than the paper currently makes, and one we would want to make deliberately.

2. **Is 15% of peak-WSE error being roughness-attributable what you would expect for a 30 m model of this event?** Calibrated, roughness accounts for 0.107 m of a 0.72 m discrepancy (Table 9); a global scale within defensible values accounts for 0.029 m. *If that is unsurprising*, the paper's framing — measure what a parameter can explain, then report it — is the contribution. *If it is far lower than expected*, the forcing or the DEM deserves scrutiny before any further calibration work, because it would mean the error budget is dominated by something we have not looked at.

3. **Does the downstream reach's failure to drain match known behaviour of this mesh and outlet configuration?** Thirty-seven of 108 surveyed marks sit in cells that fill monotonically for 72 hours with zero recession, ending +7.1 m above every surveyed value while the upstream marks are already draining (Table 6). We have excluded them from calibration and report the defect as a finding. We have ruled out a missing or misplaced exit (Section 7.1), so the question is whether this is understood behaviour, whether the cause is channel bathymetry, outlet capacity, or a domain that the event simply overtopped, and who owns it — because the bias is a smooth gradient across the domain, so it is contaminating the middle band as well, not only the reach we excluded.

4. **Is a 20% error in antecedent water storage plausible at event hour 29?** Scaling the initial state down by 20% improves peak-WSE MAE by 0.075 m (Section 8) — more than the entire physically defensible roughness range, from a perturbation that seems far easier to defend given MRMS rainfall uncertainty over a 29-hour spin-up. *If a 20% water-balance error is plausible*, the next study is the forcing and the initial state, not the friction, and we would want to know whether the rainfall product or the antecedent condition is the more likely source. *If it is not plausible*, then something else is producing a systematic high bias and the scan has found a symptom rather than a cause.

5. **Is the calibrated field itself acceptable?** We can report a field that fits better than any global scaling, but it sits 9.2 prior standard deviations from the NLCD lookup — raising developed-open by 80% and pasture by 55% above their table values, holding developed-medium at its lower bound ($n = 0.036$, smoother than concrete), and cutting woody wetland to $n = 0.035$, some 4.2σ below its lookup value (Section 7.4). We would rather be told this is uninterpretable than publish it as a calibrated roughness map. The specific judgment we need is which classes have credible ranges, as opposed to one width applied to all fifteen: the spectrum says the marks inform developed-medium (a 36% uncertainty reduction) and four others weakly, and seven classes not at all, so those seven should be frozen on physical grounds rather than left for an optimizer to move.

Open issues and current positions, in order of importance

[Temporary section, removed before submission. These are the open issues, in the order they matter to the paper, tagged with the team each belongs to: science (Gautam, Donghui, and colleagues), numerics (Matt, Jed), engineering (Jeff). Each states what we found, what we did, and the position we are proceeding from — we are not blocked on any of them. Nomenclature: "WSE" is water-surface elevation (bed elevation plus water depth), which is what a stage gauge reports once put on the NAVD88 datum; "cell bed" is the single mean ground elevation our 30 m mesh stores for a cell; "HWM" is a USGS-surveyed high-water mark, the peak WSE reached at a point during the event.]

1. **[Science] The real-data calibration stands on high-water marks, not stage.** The gauges are unusable at this resolution: modelled WSE sits 1–11 m above observed stage with a time-varying offset, because a 30 m cell stores the floodplain's mean elevation while the gauge hangs in the incised channel below it (Section 7.1 has the full account and the mark QC). *Position:* peak WSE at the 108 QC-passed marks is the primary observable. Gauge stage stays available as a secondary misfit restricted to times when model and observation are both above the cell bed, and flood-extent agreement is a candidate third; neither is needed for the headline number.

2. **[Science + numerics] Which parameterization the data actually supports.** The evidence: 238 gauge observations fit essentially perfectly while a per-cell field is 45% wrong; a class-parameterized twin at the real 13-gauge geometry (short window) drove the objective down 128× with class values still 66% wrong; the same experiment at the 108 HWM locations identifies the classes covering 83% of the domain (Section 6.1). The draft's counting argument ("overdetermined 18×") is indefensible and is gone. Three further measurements on the production configuration settle the question quantitatively: the uniform-α scan bounds the whole roughness knob at 0.08 m of a 0.72 m error; the half-domain scans show the modes are not separable, each half alone making the fit worse (Section 7.2); and one BLMVM step with all fifteen classes free drives 69.7% of the domain onto or against a bound, with developed-low rising 2.6× while developed-medium falls to the floor — a compensating pair of interleaved classes that is nearly null-space for a path-integrated observable. *Position:* the identifiability study — which classes are observable from which networks and windows — is a paper contribution in its own right. The production parameterization is *not* all fifteen NLCD classes: the observable supports roughly one degree of freedom, so the production run calibrates one to three parameters with the remainder frozen at the NLCD prior, and reports the fifteen-class run as the control that shows why. Freezing weakly-constrained parameters is the structural fix; leaning on the Tikhonov weight to suppress them after the fact is not.

3. **[Numerics] Inner tolerances and gradient fidelity.** Resolved by measurement: loose Krylov tolerances were a false economy (tightening $10^{-2} \to 10^{-3}$ is *faster*; the converged setting `ksp` 10^{-4} / `snes` 10^{-5} reproduces the fully converged 72-hour answer to every printed digit at ∼25% extra cost), so campaigns now run at converged tolerances and the verified gradients are the production gradients (Section 3.2). The long-window FD-gate question is the objective's nonsmoothness, not tolerance (Section 7.1). *Open:* principled handling of the remaining inexactness — forcing terms tied to line-search decrease, and nonsmooth-aware stopping for the quasi-Newton outer loop.

4. **[Science] Outflow boundary at zero velocity.** The critical-outflow condition switches abruptly at zero normal velocity: below it the boundary acts as a wall, above it it imposes critical outflow, so the flux jumps by roughly $gh^2/2$ as a boundary cell crosses stagnation, which pins the implicit solver (isolated by controls: a reflecting outlet completes the identical run; 5× stronger friction regularization changes nothing; tolerance relief is a treadmill because the pin level follows the local flow state). We implemented the least clever standard treatment: a *transmissive* (zero-gradient) free outflow — ghost state equal to the interior state, as in ANUGA and Clawpack — continuous by construction, differentiated exactly through host and device Jacobians. It cures the stall: the failing control completes with zero failures and two-iteration Newton solves, and a 72-hour rain-forced forward completes 259,200 implicit solves with zero failures. It is opt-in (`type: free-outflow`); the original remains the default. Caveats: exactly non-reflecting only for supercritical outflow; does not prevent inflow if the

interior velocity points inward. *Position:* transmissive outflow for the production runs, with a Froude-limited or blended form as the fallback if those caveats bite on Buffalo Bayou.

5. **[Science] Bed-friction regularization.** The friction term $\tau_b = gn^2h^{-7/3}\mathbf{q}\|\mathbf{q}\|$ switched off below the depth cutoff makes the equations jump as a cell crosses it; the jump grows like n^2, invisible in the twins but disabling at land-cover roughness ($\bar{n} = 0.10$). We now evaluate the friction with the ANUGA-regularized velocity, so drag fades smoothly to zero with depth. Off by default; runs use $h_{\text{anuga}} = 0.001\,\text{m}$, the smallest value that cures the stall; $0.003\,\text{m}$ is indistinguishable; $0.01\,\text{m}$ is unstable (mechanism conjectured: weaker damping lets an iterate reach $h < 0$). *Position:* keep the ANUGA form at $h_{\text{anuga}} = 0.001\,\text{m}$. The value is an empirical choice, not a physical one, and the sensitivity above is the whole of our evidence for it.

6. **[Numerics] Implicit solver headroom: bigger steps and better preconditioning.** Backward Euler with the assembled Jacobian carries the 2.93M-cell mesh at $\Delta t = 1\,\text{s}$ at about two Newton iterations per step on the spun-up production window (four to five in cold early transients) with `pbjacobi`-preconditioned GMRES; at $\Delta t = 5\,\text{s}$ the very first Newton *linear* solve diverges (300 GMRES iterations, `pbjacobi`) — a preconditioner failure before nonlinearity even enters. A wall-clock ratio against *explicit stepping of the same right-hand side* is not the modest number one might expect: with the Δt-free drag the adjoint requires, forward Euler is limited by the friction rate, $\Delta t < 1/t_b$ with $t_b = gn^2h^{-4/3}|v|$, and not by the CFL. On the spun-up 30 m state (99.9% wet, median depth 6 mm, t_b up to $420\,\text{s}^{-1}$) that is $\Delta t \lesssim 2.4\,\text{ms}$ against a CFL-limited 0.25 s — a factor $\sim$400 in step count, in the implicit path's favour. Confirmed end to end on the rain-forced window: forward Euler returns a non-finite state at $\Delta t = 0.25$, 0.125 *and* 0.05 s, on configurations differing from the backward-Euler run in the integrator and step size alone, while backward Euler at $\Delta t = 1\,\text{s}$ completes cleanly. The comparison that *would* be modest — explicit stepping at 0.25 s using the operator-split friction — is a comparison against a different, non-differentiable discretization, and we note it as such rather than quoting it as this method's cost. *Open:* what unlocks $\Delta t \geq 5\,\text{s}$. The candidates are a stronger preconditioner (AMG on a reduced system, block ILU, field-split), pseudo-transient continuation or another globalization once the linear solve holds, or simply accepting more Krylov iterations; whether to keep reassembling the Jacobian every step is open alongside it.

7. **[Science] Production-run design: window, observation error, gauge trust.** Four choices remain, two of them now settled by measurement. (a) *Settled.* Which phase of Harvey to calibrate on: the per-mark crest times decide it, and they are bimodal (hours 29–42 and 60–72) with an empty gap, so no 12-hour window covers both. The window is the upstream cluster, event hours 29–41, whose 46 marks are the only population in a reach the model drains correctly (Table 6); the middle band carries the same defect at intermediate strength and so cannot serve as held-out data either, which forces cross-validation *within* the upstream band. (b) The observation error σ per observable, which sets the data-versus-prior weight directly — USGS quality codes give 0.1–0.3 m for HWMs, and we are using 0.15 m. (c) The exclusion of eight reservoir-influenced sites (Addicks/Barker and Buffalo Bayou downstream), on the grounds that gate releases are unmodelled — now backed by a measurement on the mark side: the 37 downstream marks that never crest sit in cells the model demonstrably cannot drain (monotone filling through all 72 hours, zero recession, +7.1 m above every surveyed value; Section 7.1), so scoring or calibrating against them would fold a conveyance defect into the roughness field. (d) *Superseded.* The bounds are no longer stated in physical n: the production problem optimizes $\alpha = n/n_{\text{prior}}$ with $\alpha \in [0.3, 3]$ (Section 4), which is both

a physical statement about how wrong the NLCD lookup can be and a bound that excludes the region where the implicit solve fails — a box in n did neither. The NLCD field remains the prior throughout, replacing the two-zone twins' constant prior. *Newly open, and the most consequential of the four:* the prior width σ_n, which now sets the regularization directly through $\beta = \sigma_n^{-2}$. It is not a free knob: at $\sigma_n = 0.015$ the misfit still outbids the prior on the fifteen-class problem (the prior charged 118.5 units of objective and the misfit paid 146.5), and the value that would make that step net-unfavourable is $\sigma_n < 0.0135$, while $\sigma_n = 0.010$ leaves the dominant direction with an optimum at $\alpha \approx 0.88$ — a well-posed problem worth almost nothing. That tension is the result, not a tuning difficulty: within a defensible prior the observable cannot support a defensible roughness field. How wrong the NLCD lookup can plausibly be is a question for the domain team, and it now has a number attached to it.

8. **[Numerics + engineering] Upstreaming the PETSc-side pieces.** The device-resident assembled-Jacobian path relies on our PETSc branch: blocked-COO assembly for BAIJ matrices on GPU (`MatCOOUseBlockIndices`; `baijkokkos`/`baijcuda` classes) plus small TSAdjoint fixes. We think it is generally useful — one index pair per block is the natural COO for FV systems — and intend to upstream it as is. *Open:* whether the API needs changes before it hardens, and whether the paper's artifact claim should be scoped to the stock-PETSc AIJ path until it lands.

9. **[Engineering] Public API and CI.** The adjoint driver reaches through `private/rdycoreimpl.h` (the LETKF-driver pattern). *Open:* which calls deserve promotion to `rdycore.h` (`RDySetObservations`, `RDyAdjointSolve`, `RDyGetSensitivities` are the candidates), and whether Fortran bindings are needed at that point. The new tests need a TSAdjoint-capable PETSc and run in seconds; the CEED-only lanes correctly skip `numerics.jacobian` configs.

10. **[All] Venue and authorship.** Target journal (WRR / JAMES / GMD) and the co-author list remain to be settled.

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